VSC HPC introduction course

https://hpcleuven.github.io/HPC-intro

SCAN ME

Vlaanderen
is supercomputing

VLAAMS
SUPERCOMPUTER
CENTRUM

Innovative Computing
for A Smarter Flanders

Training material

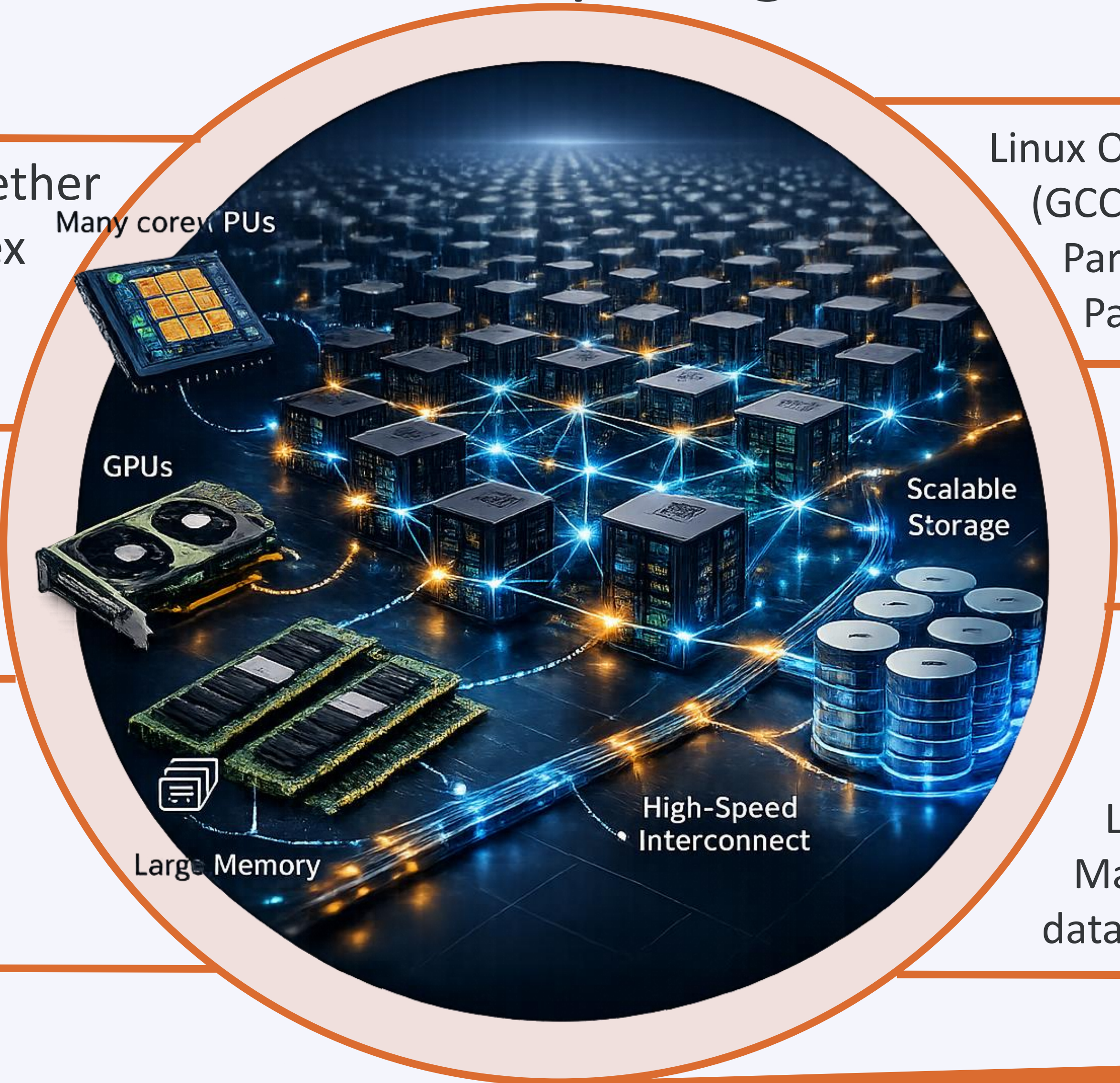
- Everything is available on Github: https://hpcleuven.github.io/HPC-intro/
- Video recordings
  - Scan the QR code
  - Recommended videos: ~2 hrs
  - Optional videos: ~1 hr

By registering and participating in VSC trainings, you accept to abide by the VSC training Code of Conduct

What is High Performance Computing?

Concept

Using many computers together to solve large-scale, complex problems faster than one computer can



Software Layer

Linux OS + system libraries
(GCC) Compilers
Parallel libraries (OpenMP, MPI, …)
Parallelized software

Hardware Layer

Hundreds of nodes
Tens of cores per node
Large memory
Fast interconnect
GPUs and Network filesystem

Applications

Genomics
Numerical (CFD) simulations
Molecular dynamics
Long-term weather predictions
Machine learning and AI
data pre-/post-processing

What is High Performance Computing?

The concept is simple: **parallelism** = employing multiple processors for a single problem

Training outline

Part 1: Basics

- ❑ VSC
- ❑ Tier-2 clusters
- ❑ Storage
- ❑ Accounting
- ❑ Login
  - Open OnDemand
  - SSH client & terminal
- ❑ Data transfer
  - FileZilla
  - rsync
  - Globus

Part 2: Hands-on

- ❑ Using Open OnDemand
- ❑ Software
- ❑ Miniconda
- ❑ Batch jobs
- ❑ Demo: try things yourself

VSC

(Vlaams Supercomputer Centrum)

VSC PARTNERSHIP

Supported by

fwo

ASSOCIATIE UNIVERSITEIT-HOGESCHOLEN LIMBURG

ASSOCIATIE UNIVERSITEIT GENT

universitaire associatie BRUSSEL

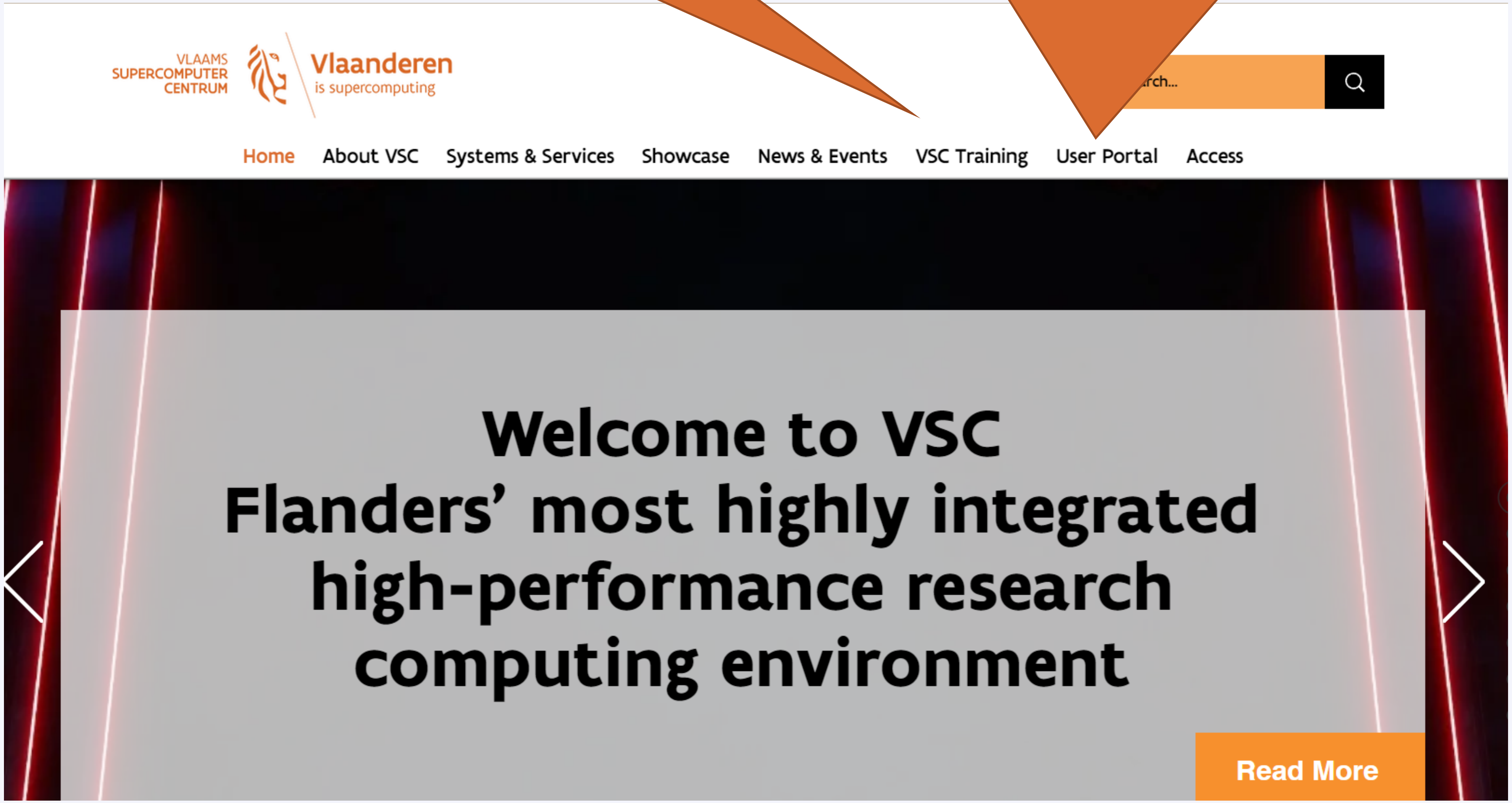
ASSOCIATIE KU LEUVEN

auha antwerp university association

Visit https://vscentrum.be

Extensive set of courses every academic year

Links to the VSC **account page** and the VSC **documentation** (answers >80% of your questions!)



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is supercomputing

Search...

Home | About VSC | Systems & Services | Showcase | News & Events | VSC Training | User Portal | Access

Welcome to the User Portal

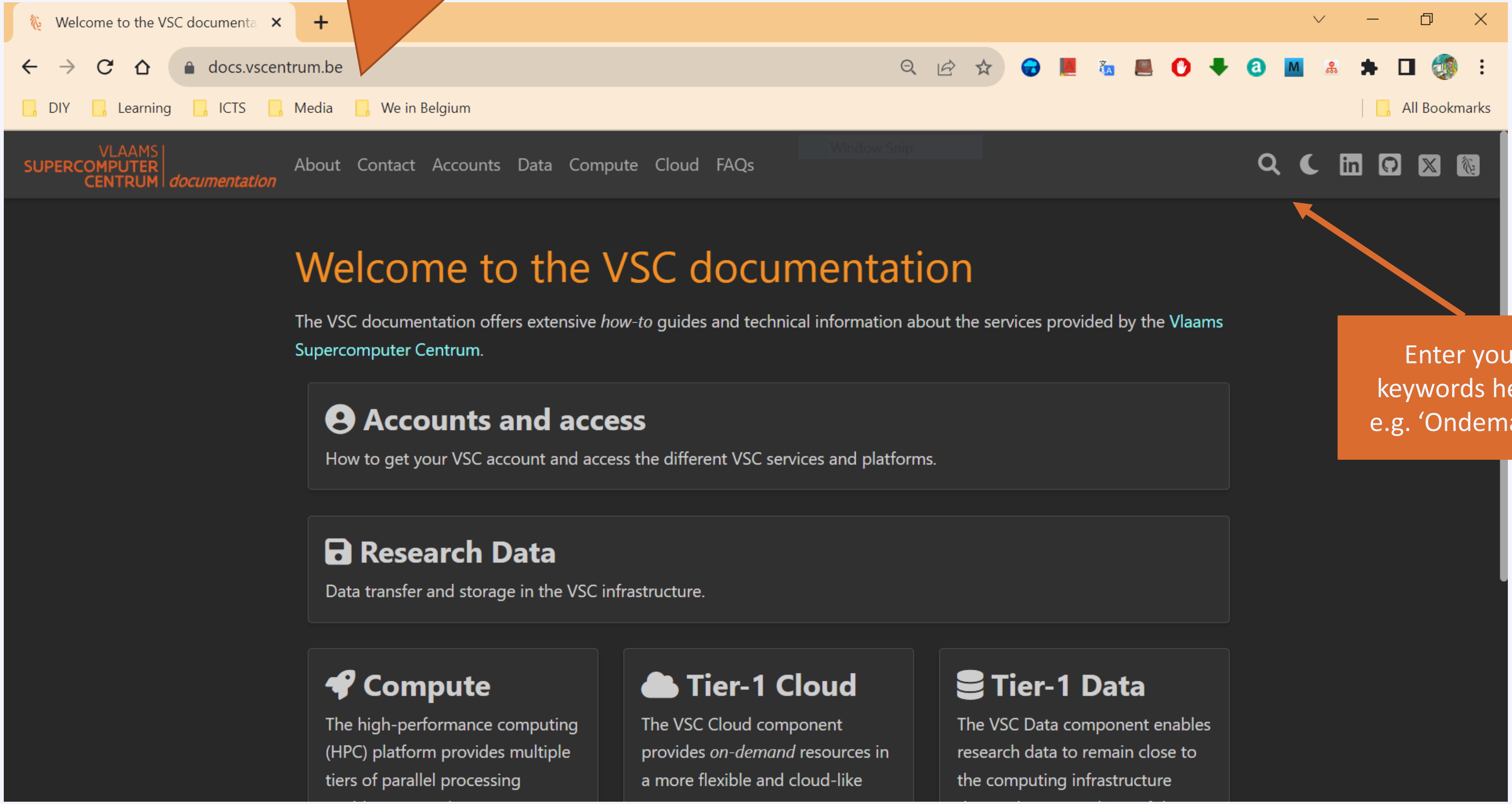
Here you can find the gateway to the User documentation of the Vlaams Supercomputer Centrum

User documentation page

Manage your VSC-account
partner institute account required

VSC account page

https://docs.vscentrum.be



Enter your keywords here; e.g. 'Ondemand'

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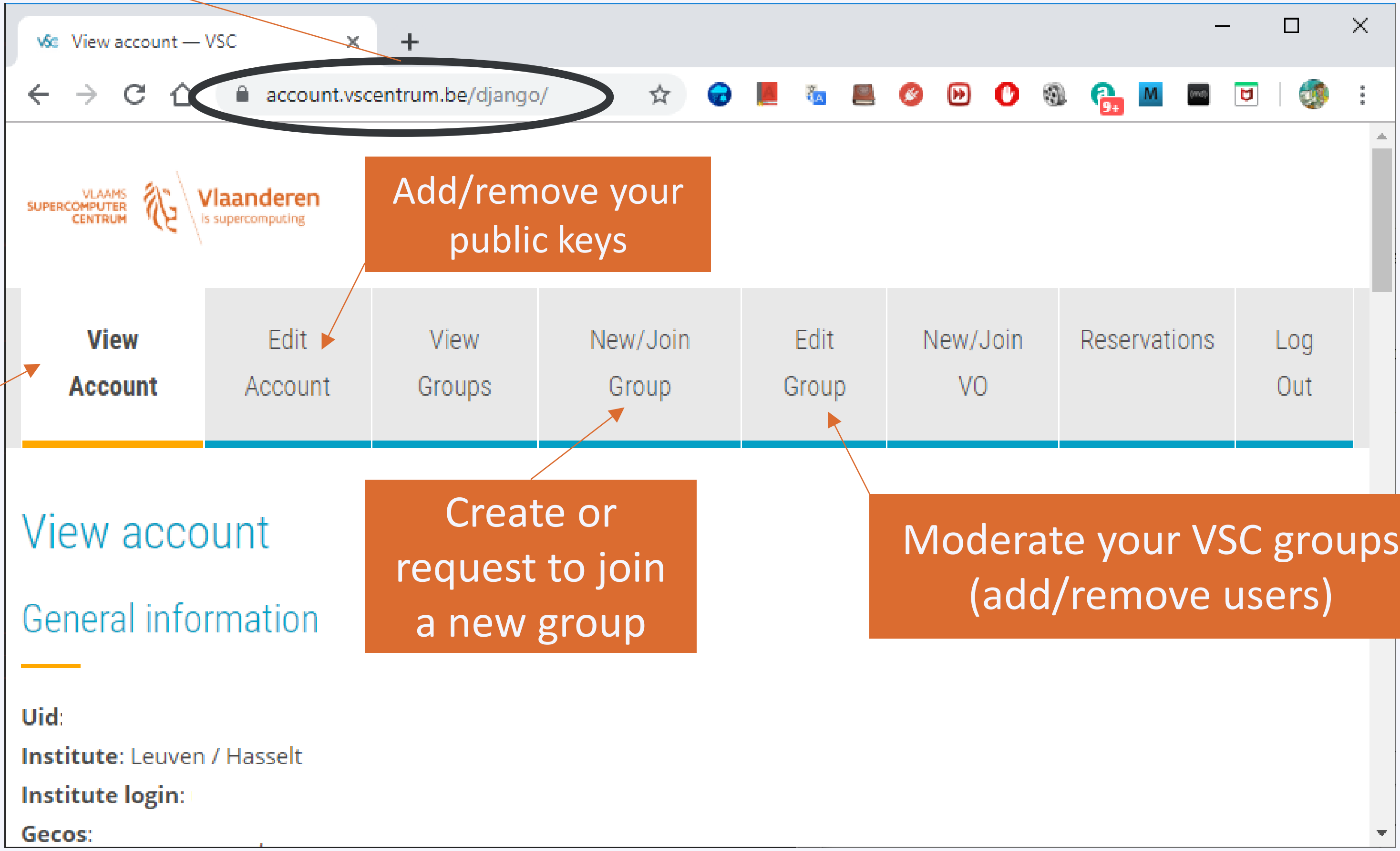
Manage your VSC account at
https://account.vscentrum.be



Add/remove your public keys

View public keys
Request account

Create or request to join a new group

Moderate your VSC groups (add/remove users)

Support and services

Basic support

- Helpdesk (hpcinfo@kuleuven.be)
- Monitoring and reporting

Application support

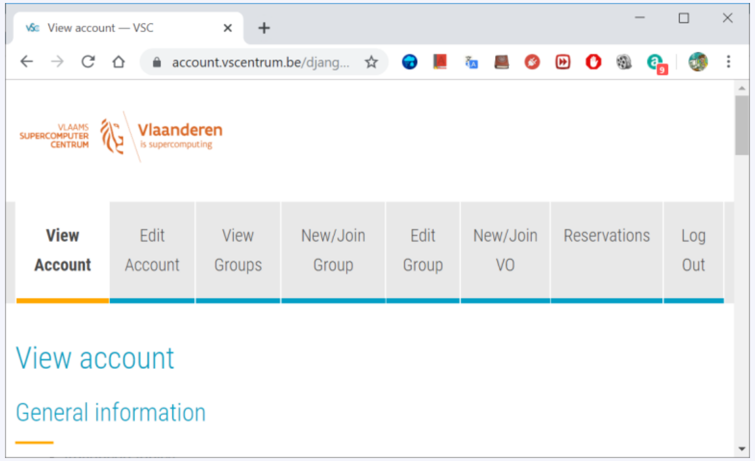
- Installation and porting
- Optimisation and debugging
- Benchmarking
- Workflows and best practices

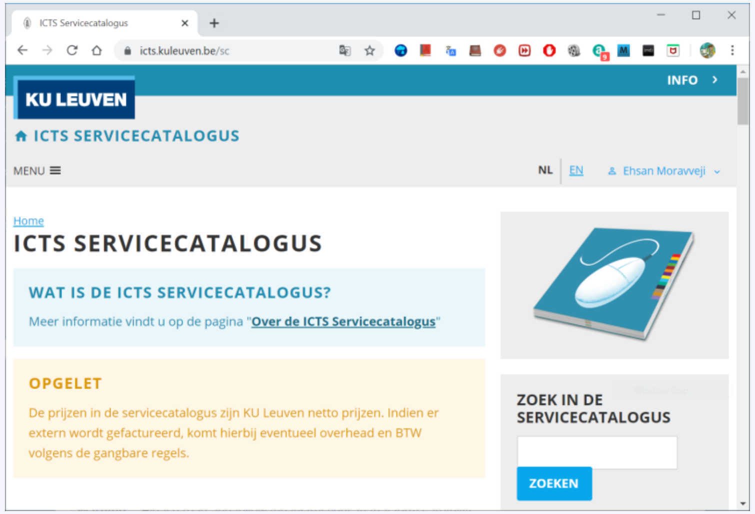
Training

- Documentation and tutorials
- Scheduled trainings and workshops
- Workshops on request
- One-to-one sessions

Become a VSC user

- (Optional) Create a secure (4096 bit) SSH key pairs
  Upload it on the account page: https://account.vscentrum.be
- You need to request a VSC account
  Normally gets processed swiftly
- Request introductory credits (2M free credits for 6 months)
- Request project credits (for supervisors and project leaders)
  You need to create a VSC group
  Add users to the group to give them access to use credits
  Fill out the request form
- Extra storage requests
  Scratch extension: free of charge
  Staging storage: 20 € per TB per year
- All service costs (compute and storage) are described in the ICTS service catalog: https://icts.kuleuven.be/sc
  Click on High Performance Computing (NL/EN)





VSC training

Infosessions:
- Containers
- Notebooks

- Introductory
- Intermediate
- Advanced
- Specialized track

Matlab (Supercalculator)
Matlab programming

Linux

HPC intro

C

C++ for scientific computing

Fortran for programmers

High Performance Python

?

Linux for HPC

Make intro

CMake intro

Linux scripting

Linux tools

Version control systems

- Python Introduction
- Python: System programming
- Scientific Python
- Python for software engineering
- Python for data science
- Python for machine learning

worker/atools

Julia good bad ugly

MPI

OpenMP

Debugging techniques

Code optimization

Stay up-to-date https://www.vscentrum.be/en/education-and-trainings

How to acknowledge VSC in publications

Why?

- a contractual obligation for the VSC
- helps VSC to secure funding
- you will benefit from it in the long run

For KU Leuven users

- add the relevant papers to the “KUL-HPC” virtual collection in Lirias

In het Nederlands

De infrastructuur en dienstverlening gebruikt in dit werk werd voorzien door het VSC (Vlaams Supercomputer Centrum), gefinancierd door het FWO en de Vlaamse overheid.

In English

The resources and services used in this work were provided by the VSC (Flemish Supercomputer Center), funded by the Research Foundation - Flanders (FWO) and the Flemish Government.

Tier-2 clusters

VSC HPC infrastructure

Tier-0

EURO
BELGIUM

PRACE

Tier-1

sofia (VUB, 2026)

Hortense
(UGent, 2022 - 2026)

Vlaanderen
is supercomputing

Tier-2

GHENT UNIVERSITY	VRIJE UNIVERSITEIT BRUSSEL	Universiteit Antwerpen	KU LEUVEN / UHASSELT
STEVIN	**HYDRA**	**LEIBNIZ/VAUGHAN**	**Genius/wICE/Mindwell**

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sofia

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Tier-2 clusters @ KU Leuven

Genius (2018 - 2026)
16 nodes | 576 cores | 72 GPUs

wICE (since September 2022)
260 nodes | 20272 cores

Mindwell (since May 2026)
55 nodes | 10560 cores

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Tier-2 cluster – Genius

Node type	CPU type	# cores	Total mem	Mem per core (MB)	Partition	Planned end of life
Skylake P100 **GPU**	Xeon Gold 6140 **4xP100** SXM2 16GB	36	**192 GB**	**5000**	gpu_p100	2027 (?)
Cascadelake V100 **GPU**	Xeon Gold 6240 **8xV100** SXM2 32GB	36	**768 GB**	**21000**	gpu_v100	2027 (?)

Remarks

- **Access:** login.hpc.kuleuven.be
- **CPU nodes:** Decommissioning: March 2026
- **GPU nodes:** Will stay until 2027
- **Current OS:** Rocky8
- **Future OS:** Rocky9 (soon)

Tier-2 cluster – wICE

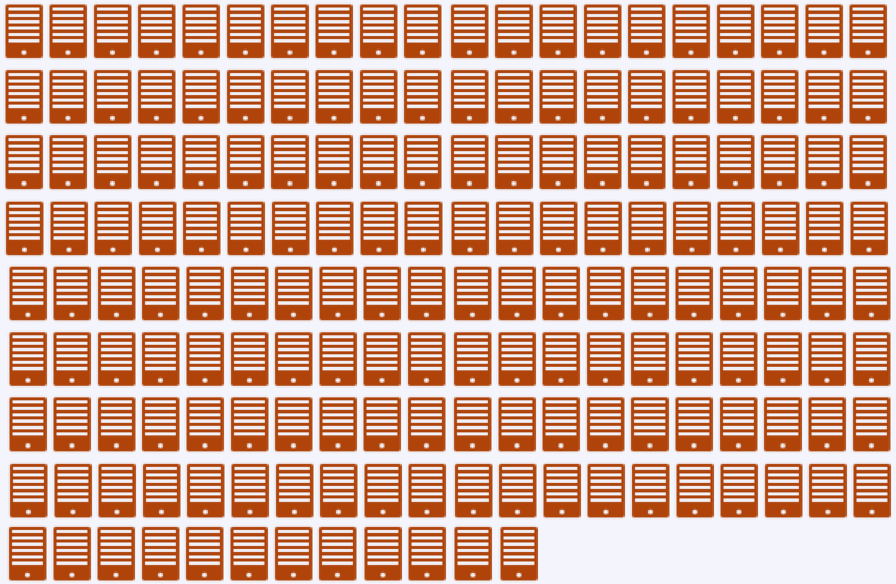
Node type	CPU type	# cores	Total mem	Mem per core (MB)	Partition
Icelake	Xeon 8360Y	72	**256 GB**	**3455**	batch_icelake
Sapphire Rapids	Xeon 8468	96	**256 GB**	**2500**	batch_ sapphirerapids
Icelake **large mem**	Xeon 8360Y	72	**2 TB**	**28000**	bigmem
Icelake **large mem**	Xeon 8360Y	72	**8 TB**	**111900**	hugemem
Icelake A100 **GPU**	Xeon 8360Y **4xA100** SXM2 80GB	72	**512 GB**	**7000**	gpu_a100
Genoa H100 **GPU**	Epyc 9334 **4xH100** SXM5 80GB	64	**768 GB**	**11700**	gpu_h100
Icelake **interactive**	Xeon 8358 **1xA100** SXM2 80GB	64	**512 GB**	**7500**	interactive

Remarks

- **OS:** Rocky 9.6 (after 18 Feb '26)
- **Access:** login.hpc.kuleuven.be
- **Interconnect:** InfiniBand HDR (100 Gb/s)
- **SSD Disks:** 960 GB
- **Interactive partition**:
  max resources: 8 cores, 1 GPU, 16hr

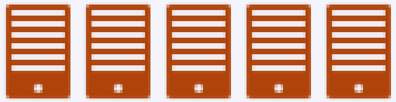
Tier-2 cluster – wICE

Thin nodes



172x IceLake 72c 256 GB

Large memory nodes



5x IceLake 72c 2048 GB

GPU nodes

4x IceLake 72c 512 GB
4 A100 SXM4 80GB

Interactive nodes

4x IceLake 64c 512 GB
1 A100 80GB

No Dedicated Login Nodes

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Tier-2 cluster – wICE extension

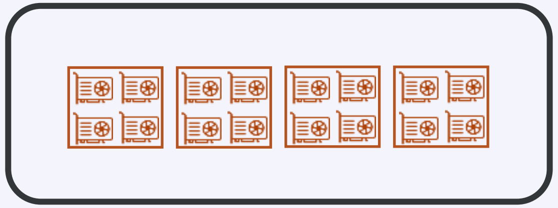
Thin nodes

68x Sapphire Rapids 96c 256 GB

Huge memory node

1x IceLake 72c 8 TB

GPU nodes



4x AMD Genoa 64c 768 GB
4x H100 SXM5 80GB

Storage

Overview of the Tier-2 storage locations

- Only you own your files (POSIX)
  Users can share folders via VSC groups
- A VSC account has 3 default storages (free of charge)
  - `$VSC_HOME`
  - `$VSC_DATA`
  - `$VSC_SCRATCH`
- You can additionally request `staging` storage
- Different storage volumes have different:
  - mount point
  - size and performance
  - use case
  - backup and maintenance policy
- For more info, see the ICTS Service Catalog (EN/NL)

Storage locations

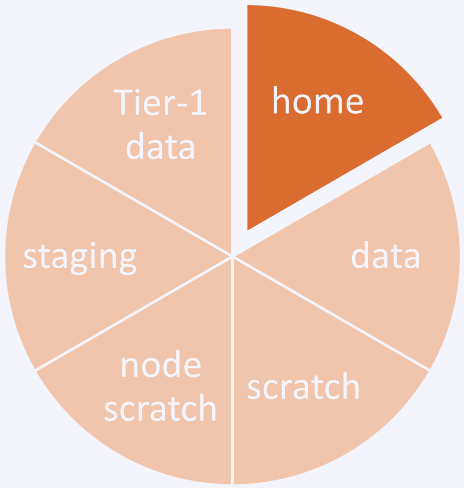
- **Avoid using `/tmp` (`$TMPDIR`) on the login nodes**
  It is only 10 GB and is needed by the operating system
  Your application can crash if using `/tmp`
- You are automatically logged into your home folder upon login. Best to immediately switch to some other storage, e.g.
  `$ cd $VSC_DATA`
- Always check your storage balance using `myquota` command

Example

```
$ myquota
file system $VSC_HOME
    Blocks: 1479M of 3072M
    Files: 12934 of 100k
file system $VSC_DATA
    Blocks: 12G of 75G
    Files: 1043k of 10000k
file system $VSC_SCRATCH
    Blocks: 15M of 1.5T
```

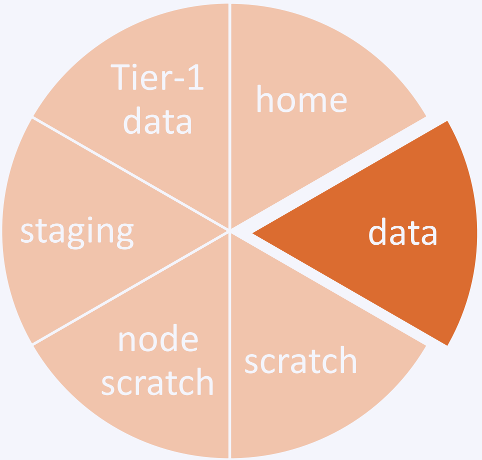
- Request form for extra storage
- More info in the service catalog

Storage locations



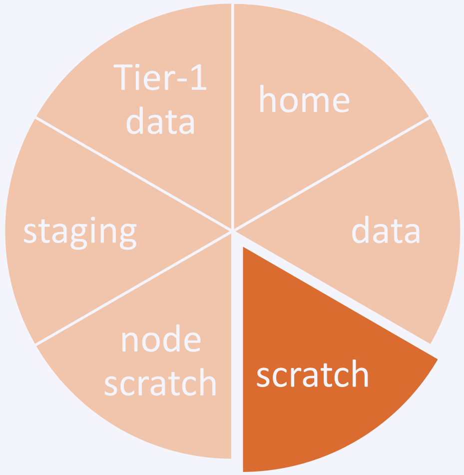
Storage	Home folder
Env. variable	`$VSC_HOME`
Filesystem type	NFS
Access	Global
Backup	Hourly, daily, weekly (until last month) inside `.snapshot` folders
Default quotum	3 GB
Quotum increase	Not possible
Usage	Only storing SSH keys, config files, …
Remarks	- Stay away from using it - Can easily overflow: + Your jobs may crash + Login issues

Storage locations



Storage	Data folder
Env. variable	`$VSC_DATA`
Filesystem type	NFS
Access	Global
Backup	Hourly, daily, weekly (until last month) inside `.snapshot` folders
Default quotum	75 GB
Quotum increase	On purchase
Usage	Your data, code, software, results
Remarks	- Permanent storage for initial/final results - Not optimal for intensive or parallel I/O

Storage locations



The Lustre filesystem is nearly full. Momentarily, we don't extend user quota's with more than 1TB.

Storage	Scratch folder
Env. variable	`$VSC_SCRATCH`
Filesystem type	Lustre & GPFS
Access	Local
Backup	Files get deleted 30 days after last access
Default quotum	500 GB for both
Quotum increase	Free
Usage	Intensive, parallel I/O, temporary files
Remarks	- Recommended storage for all jobs - Copy scratch files to VSC_DATA or local storage after jobs are done - Deleted files cannot be recovered - Avoid folders with >10,000 files

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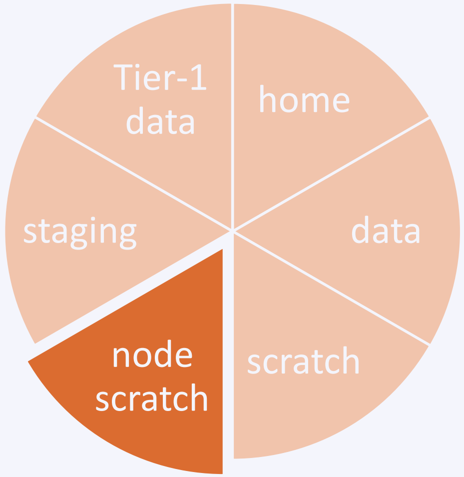
Storage locations



Storage	Node scratch folder
Env. variable	`$VSC_SCRATCH_NODE`
Filesystem type	Ext4
Access	On compute node, only at runtime
Backup	None
Default quotum	591 GB
Extension	Read about beeOND
Usage	Temporary storage at runtime
Remarks	- Fastest I/O, attached to the node - Is cleaned after job terminates - Copy the data to your home, scratch, or staging before job ends

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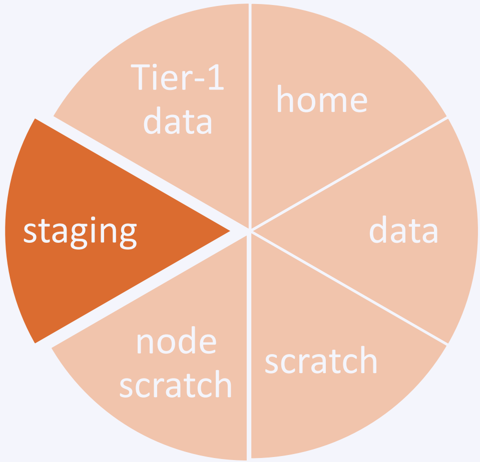
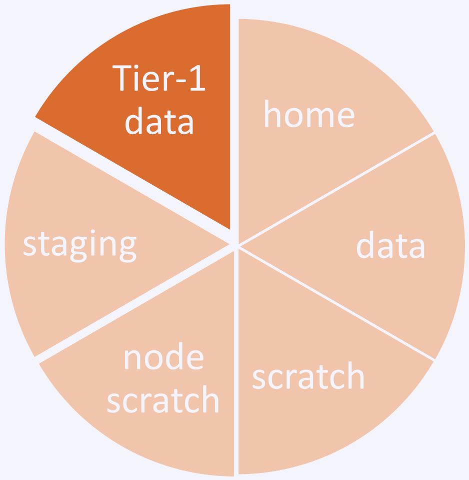
Storage locations



The Lustre filesystem is nearly full.
Momentarily, we don't create or extend users' staging quota requests.

Storage	Staging folder
Path	`/staging/leuven/stg_000XX`
Filesystem type	Lustre
Access	On demand, only Tier-2@KUL
Backup	None
Default quotum	None
Extension	On purchase, from 1 TB
Usage	Permanent; share with a group
Remarks	- Accessible from login/compute nodes - Fast, parallel I/O

Storage locations



- VSC offers Tier-1 Data as a large storage service
- Only for active data (no archiving)
- Supports metadata (search & discover)
- Share within VSC (permission management)
- Share outside VSC (via Globus)
- Stage-in/stage-out data workflow
- Free of charge for academic use
- Starting volume is 5 TB
- Submit a proposal to get access
  You can apply at any time

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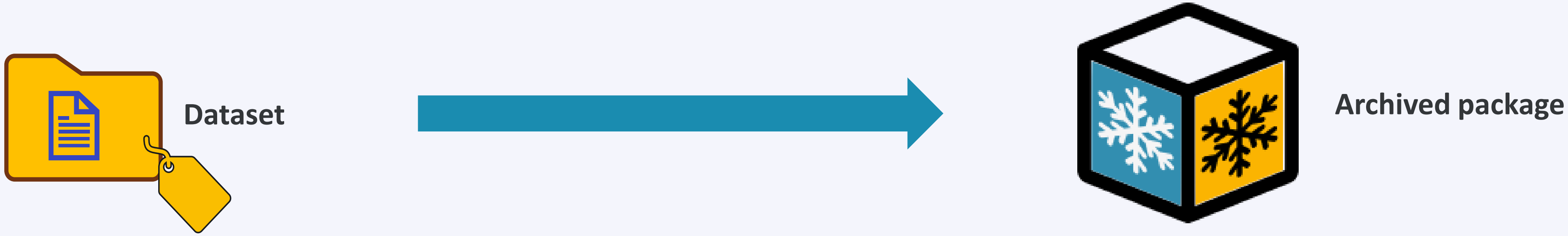
Extra (KU Leuven) storage locations



ManGO

- In production
- For **active** data
- Data should stage in/out
- Has backup
- See the ICTS service catalog

FriGO

- In development
- For **inactive** data
- Storage until 10 years
- Has backup
- See the ICTS service catalog

Also see the KU Leuven storage guide for even more storage possibilities

ManGO

Central, secure,
highly available

STORE

Apply rich metadata,
enforce conventions,
standardize

DESCRIBE

AUTOMATE

Implement rule-based
policies & workflows

SHARE

Collaborate securely
& efficiently

Compute credits

Tier-2 credit system

What?

- Credits are a measure for compute resources
- Mandatory for accessing these resources
- Traceability and accountability

Why?

- Incentive to optimize your code and workflow

How?

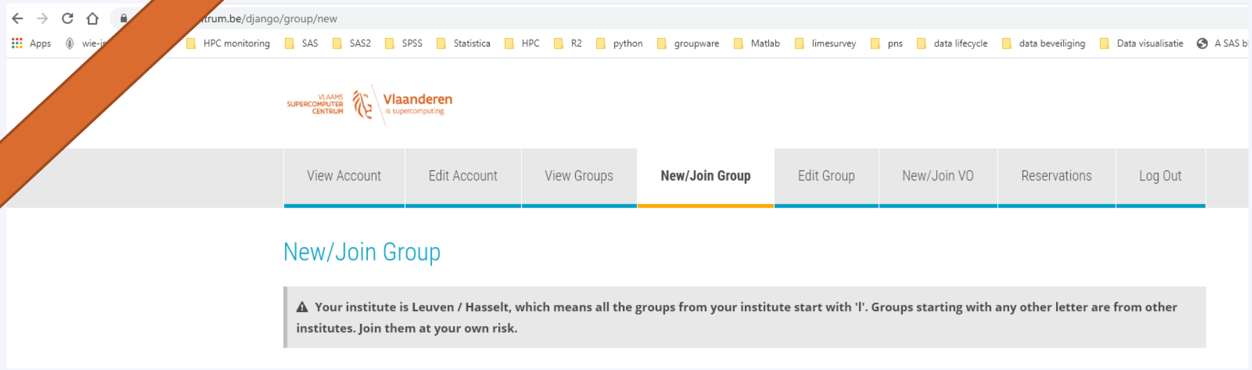
- 1 million credits = 3.5 EUR
- Pay-as-you-go model
- Credit rates on the right = for fully allocated nodes (if you request less cores, you also get billed less)
- See the [ICTS service catalog] for more info

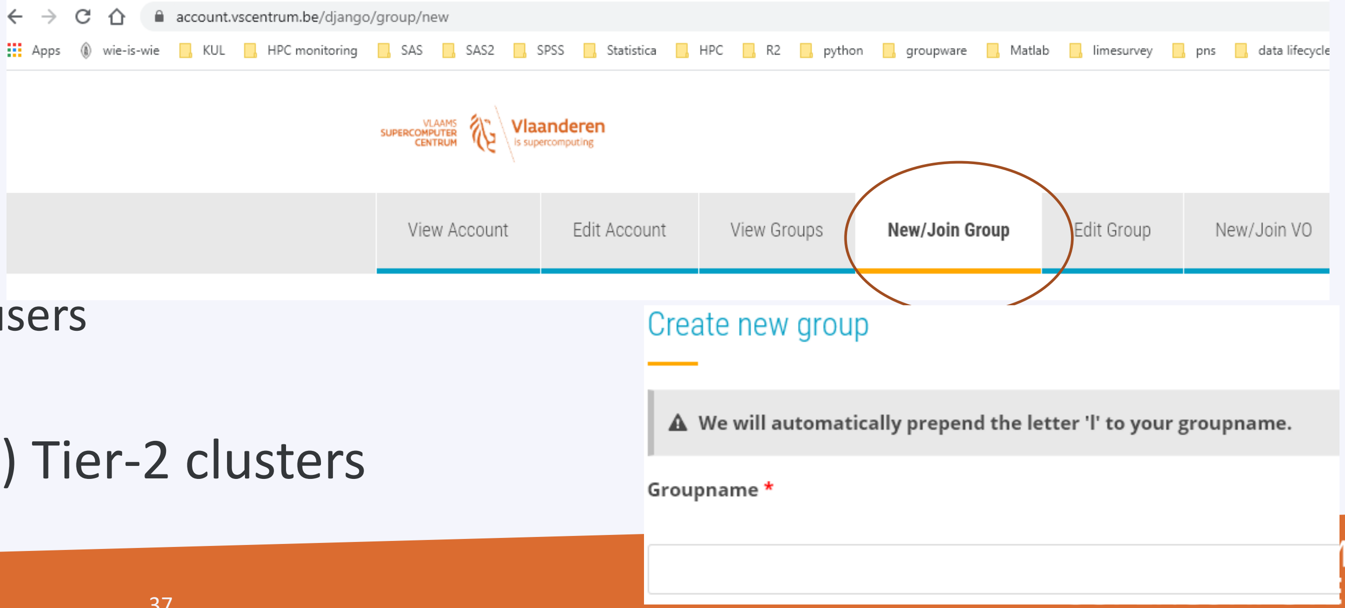
Node types	Credits per node-hour
Genius Skylake 4 GPUs	20 000
Genius Cascadelake 8 GPUs	39 900
wICE IceLake thin node	11 000
wICE IceLake Large Memory	19 000
wICE IceLake GPU node	45 000
wICE IceLake interactive node	Free
wICE IceLake hugemem node	80 000
wICE Sapphire Rapids	20 000
wICE AMD Zen-4 4 GPUs	150 000

Tier-2 credit system – how to request credits

- Request your introduction credits
  - Every new user can request them once
  - Free of charge
  - 6 months
- Request permission to join a group
- Request project credits
  - Create group on the VSC account page
  - Start the name preferably with **p_**
  - Your group will start with **lp_**
  - Creator of the group can add and remove users
- Tier-2 credits can be used on all (Leuven) Tier-2 clusters

Check the [ICTS service catalog]





Tier-2 credit system – how

- Submit your jobs with a suitable credit account

  #SBATCH –A lp_myproject

  sbatch -A lp_myproject … myjob.slurm

- Before submitting the job, Slurm checks if the account has enough available credits for the job (in addition to any other already-running jobs)

- Once a job start running, it starts consuming credits from the account

Tier-2 credit system – manage your credits

- We provide various `sam`-commands to facilitate credit related queries

Command	Purpose
`sam-balance [-A account]`	List active projects and available credits

```
tier2-p-login-3$ sam-balance -A lp_hpcinfo
ID       Name                    Balance          Reserved         Available
======== ======================= ================ ================ ================
96141    lp_hpcinfo              9782636          0                9782636
tier2-p-login-3$
```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Tier-2 credit system – manage your credits

- We provide various `sam`-commands to facilitate credit related queries

Command	Purpose
`sam-list-allocations -A <account-name>`	List the credit deposits and refunds (if any)

```
tier2-p-login-3$ sam-list-allocations -A lp_hpcinfo
AllocID    AccountID Account           Timestamp              Credits
========== ========= ================= ====================== ================
80745      96141     lp_hpcinfo        2023-02-14T10:40:50    2000000
203565     96141     lp_hpcinfo        2023-04-17T16:42:26    2000000
227730     96141     lp_hpcinfo        2023-05-03T09:10:29    2000000
313474     96141     lp_hpcinfo        2023-06-19T17:33:59    8410000
tier2-p-login-3$
```

Tier-2 credit system – manage your credits

- We provide various *sam*-commands to facilitate credit related queries

Command	Purpose
`sam-quote sbatch [sbatch args] [myjob.slurm]`	Estimates the (max) amount of credits a job will use

```
tier2-p-login-3$ sam-quote sbatch --cluster=wice hello.slurm
10980
tier2-p-login-3$
```


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Tier-2 credit system – manage your credits

- We provide various `sam`-commands to facilitate credit related queries

Command	Purpose
`sam-statement -A <account-name> -s <start-date> -e <end-date>`	Overview of credits used by each job in some time window

```
tier2-p-login-3$  sam-statement -A lp_hpcinfo -s 2023-09-01 -e 2023-10-25
################################################################################
#
# Includes Account=lp_hpcinfo
# Generated on Mon Oct 30 11:01:29 2023.
# Reporting fund activity from 2023-09-01 to 2023-10-25
#
################################################################################

---------------------------------------- --------------------
Credits deposited in the given period:                      0
Credits refunded in the given period:                       0
Credits consumed in the given period:                       0
---------------------------------------- --------------------

JobID      Cluster  Account               User     Partition   Credits
========== ======== ===================== ======== =========== ===============
60739746   wice     lp_hpcinfo            vsc30446 batch       0
60740123   wice     lp_hpcinfo            vsc30446 batch       0
60740125   wice     lp_hpcinfo            vsc30446 batch       0
60746730   wice     lp_hpcinfo            vsc30446 interactive 0
60773547   wice     lp_hpcinfo            vsc30446 interactive 0
60774141   wice     lp_hpcinfo            vsc30446 batch       0
60774160   wice     lp_hpcinfo            vsc30446 batch       0
60783857   wice     lp_hpcinfo            vsc30446 interactive 0
```

Tier-2 credit system – manage your credits

- We provide various sam-commands to facilitate credit related queries

Command	Purpose
sam-balance	Lists active projects and available credits
sam-list-allocations -A <account-name>	Lists the credit deposits and refunds (if any)
sam-quote sbatch [sbatch args] [myjob.slurm]	Estimates the (max) number of credits a job will use
sam-statement -A <account-name> -s <start-date> -e <end-date>	Overview of credits used by each job in some time window

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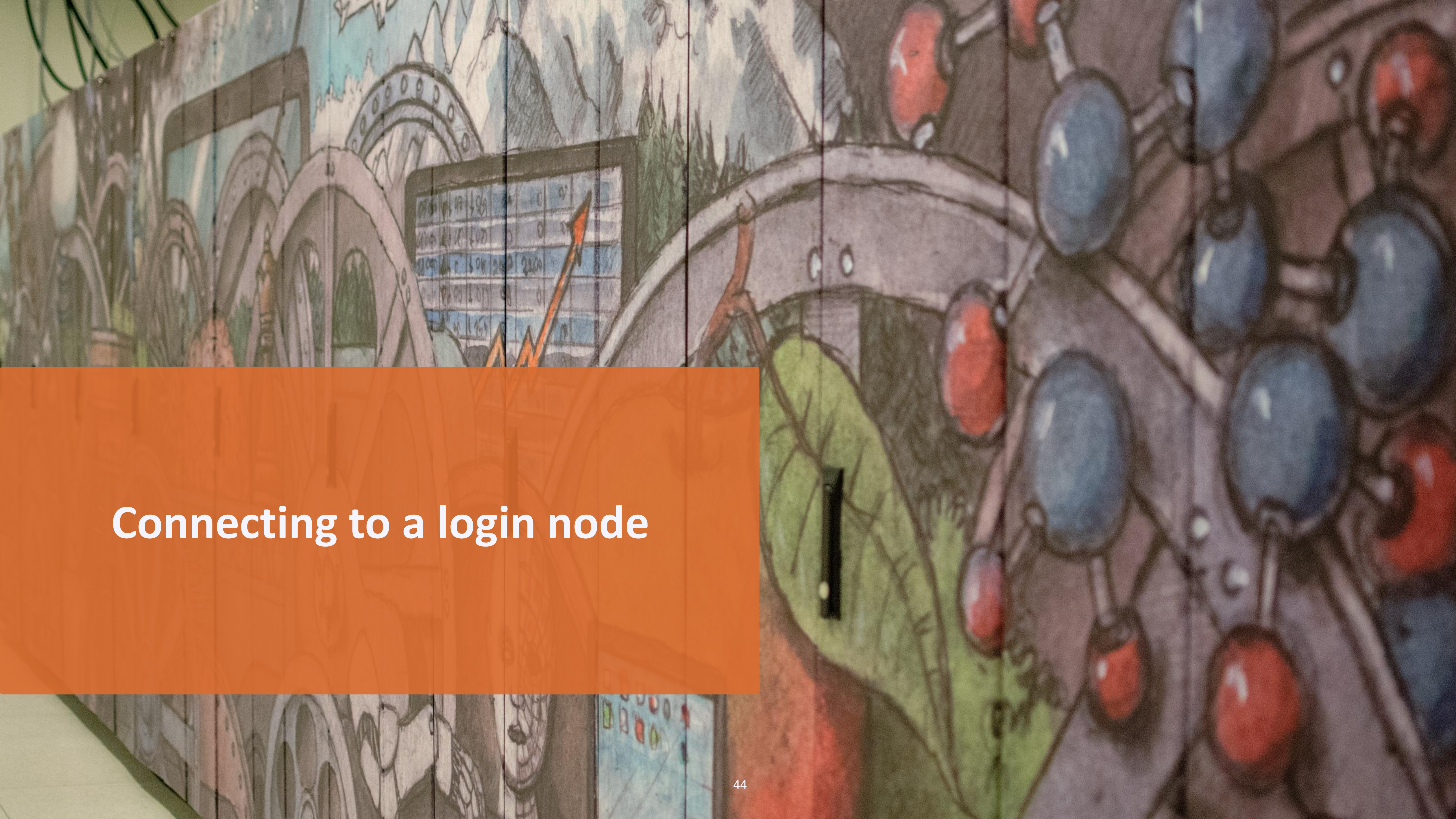
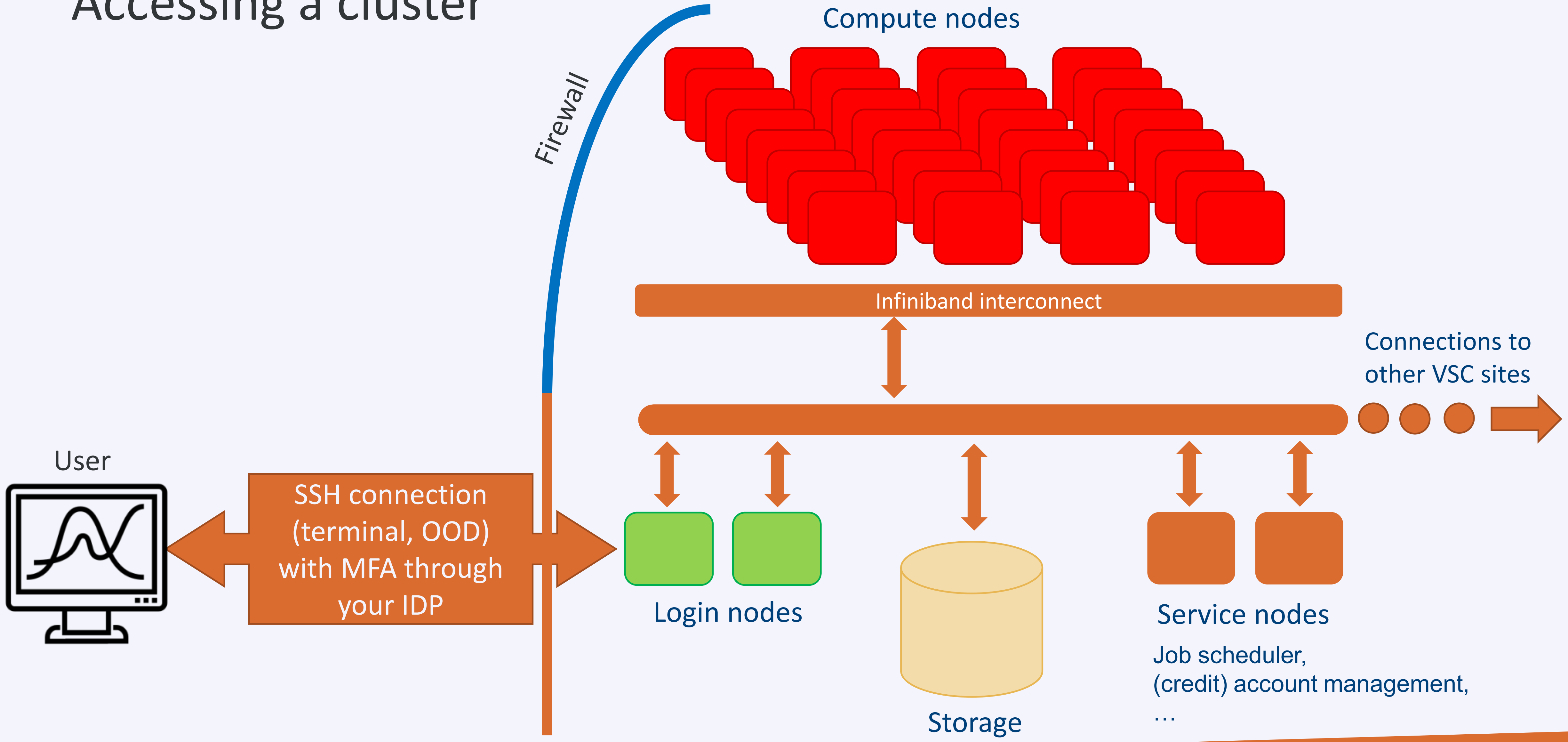


Connecting to a login node

Accessing a cluster



How to login?

Open OnDemand

- Web-based login via your browser
- https://ondemand.hpc.kuleuven.be
- Multi-Factor Authentication (MFA): Login through your institute's IDP
- No SSH key needed

SSH client

- Windows: PuTTY, MobaXterm, …
  MacOS/Linux: any terminal
- Optional: for VSC clusters *not* hosted at KU Leuven you need an (open)SSH key pair (upload the public key to the VSC account page and wait ≥ 30mins)
- Hosts: login.hpc.kuleuven.be:22
  Recommended to use an SSH agent
- Can open GUIs if X11 forwarding is enabled

Using the login nodes

Tips

- ❑ To develop code
- ❑ To check your storage and credit balance
- ❑ To manage jobs (submit, check status, debug, resubmit, …)
- ❑ To move data around
  - within VSC: use data, scratch, staging
  - outside VSC: copy/sync from/to your local storage (e.g. Globus)
- ❑ To pre- or postprocess your data
- ❑ To visualize your data
- ❑ To share files and folders

Warning

Do NOT execute CPU- or memory-intensive tasks here

Login nodes are shared resources

Submit jobs instead (you can e.g. compile software on compute nodes via an interactive job)

Use `slurmtop` to see how busy the cluster is

Connecting via a terminal

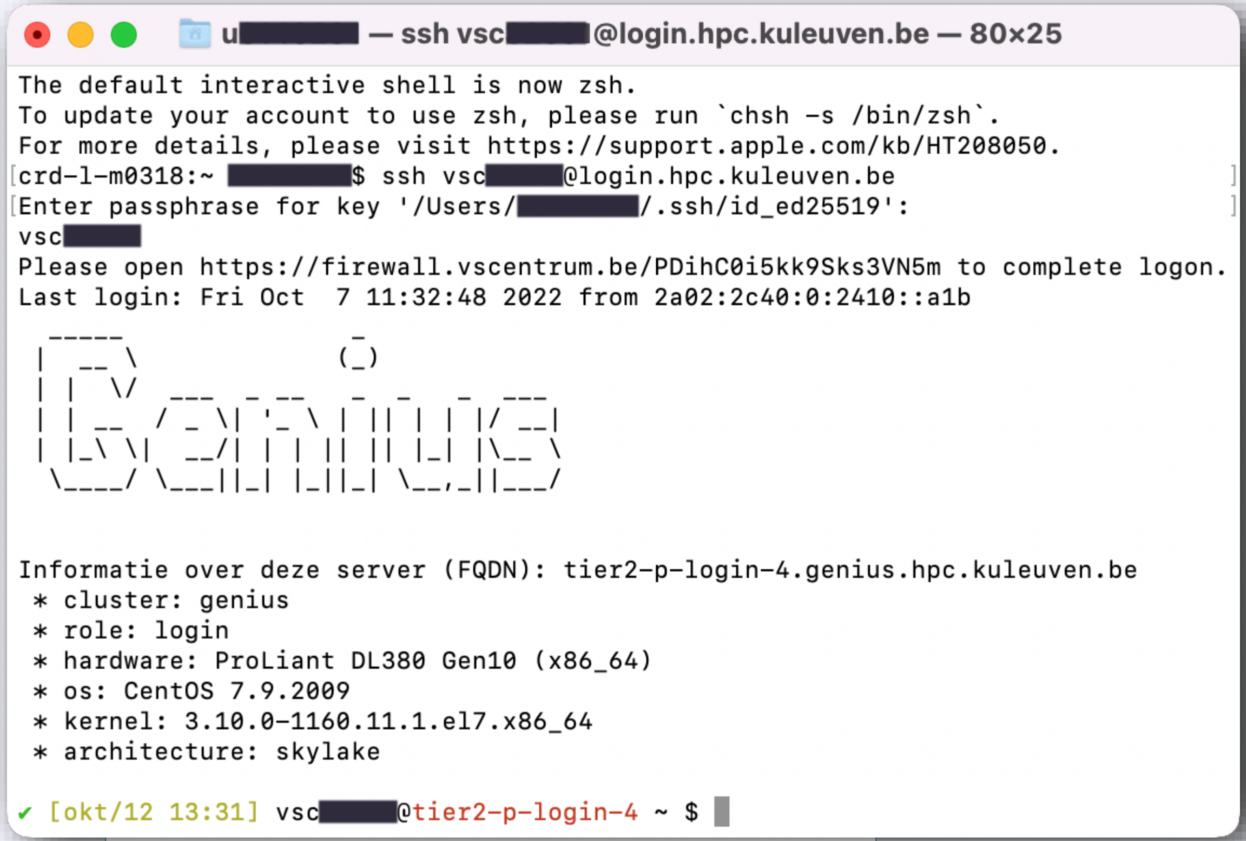
(Linux and Mac)

Hostname:

`login.hpc.kuleuven.be`

- Use ssh to connect:
  `$ ssh -A vscXXXXX@<host_name>`
- If key not found:
  `$ ssh `**`-i </path/to/keyfile>`**` ...`

If asked for **password**, please stop connecting and contact support, otherwise after a few attempts you will be blocked for 24h.



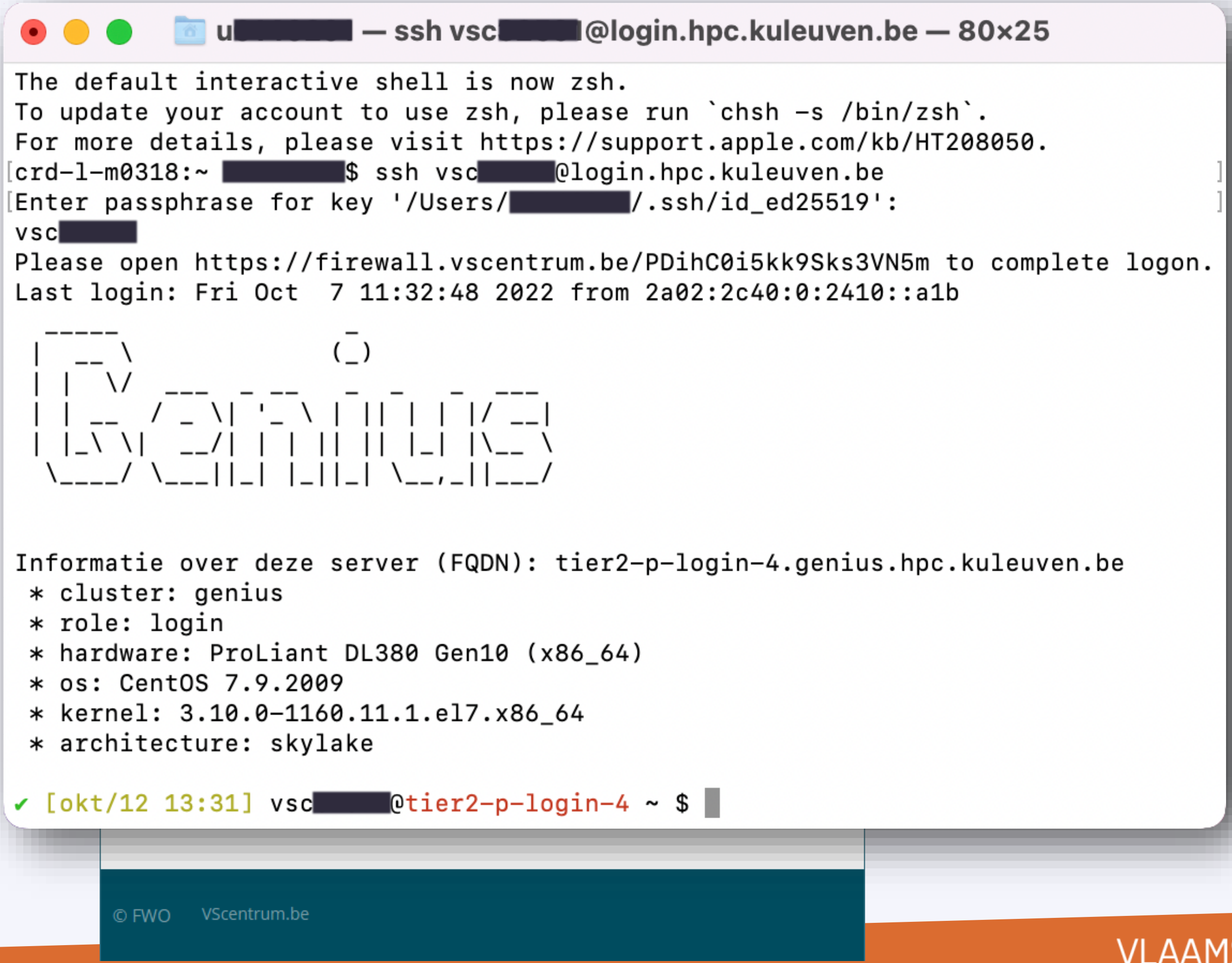
Connecting via a terminal

(Linux and Mac)

With an SSH agent

- Check your SSH agent
  Is your SSH key found?
  `$ ssh-add -l`
- If your SSH agent is not running:
  `$ eval $(ssh-agent)`
- If your key is not found,
  add it to the agent:
  `$ ssh-add </path/to/keyfile>`
- Use ssh to connect:
  `$ ssh -A vscXXXXX@<hostname>`

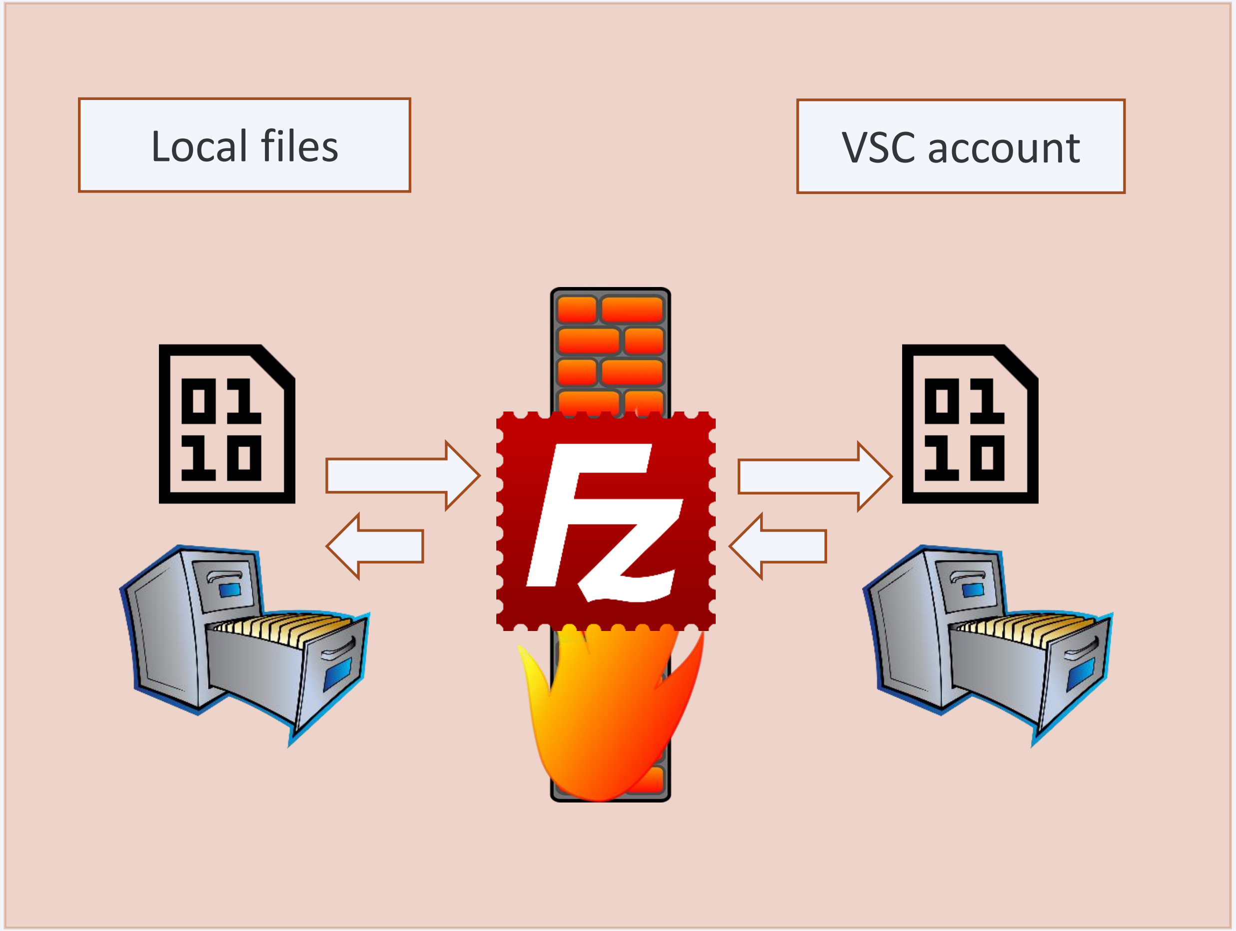
Hostname:
`login.hpc.kuleuven.be`



VLAAMS SUPERCOMPUTER CENTRUM

Data transfer

File transfer tools

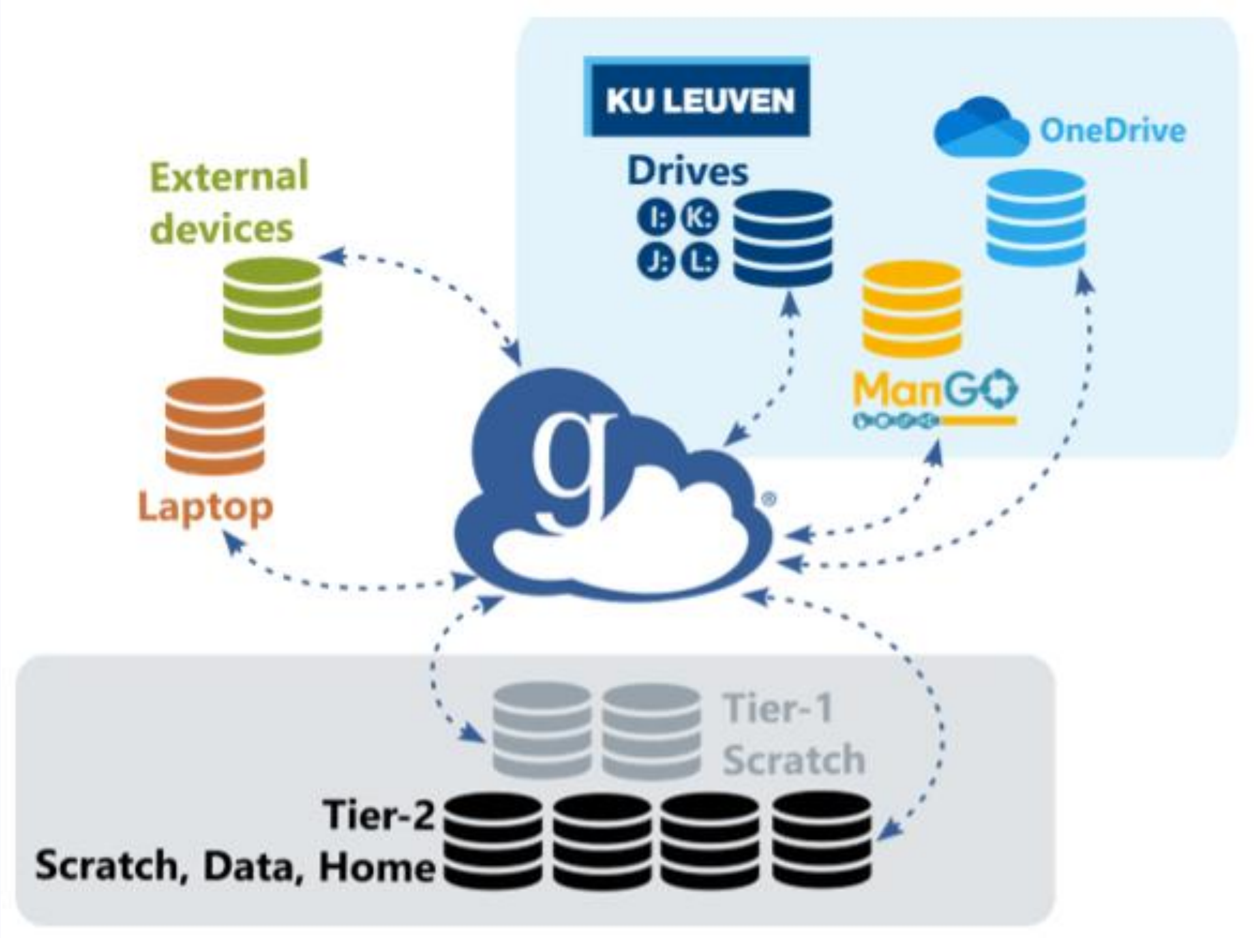


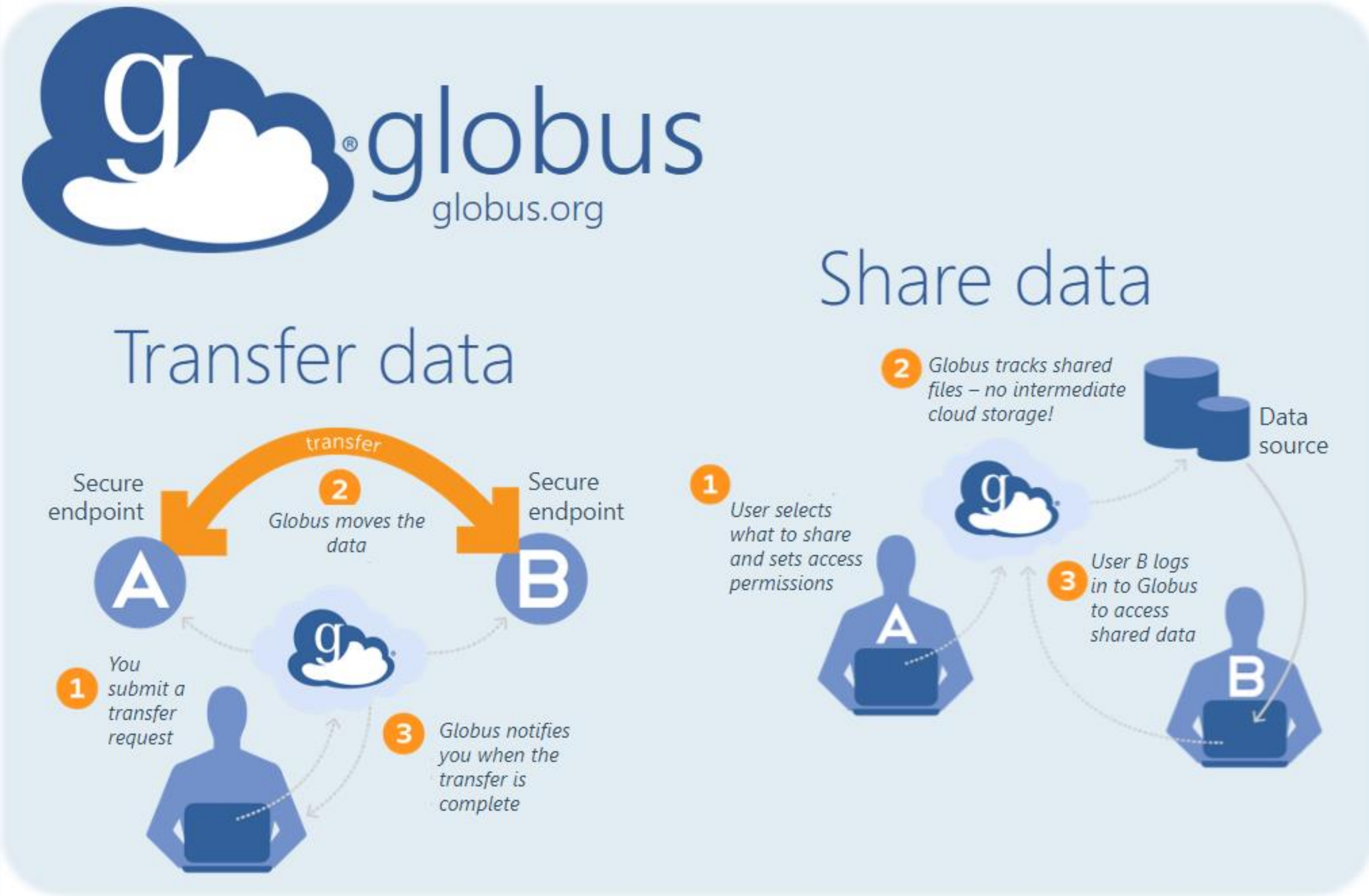
Application	OS
FileZilla	Windows, Linux, Mac
WinSCP	Windows
rsync, scp	Linux, Mac
Globus	Window, Linux, Mac

File transfer with Globus

- ❑ Web interface for transferring, sharing and searching within data
- ❑ Schedule and resume (large) transfers
- ❑ Workflow:
  Define “endpoint” on the source
  Define “endpoint” on the destination
  Transfer between endpoints
- ❑ Endpoints on all VSC sites:
  all Tier-2 clusters: home, data, scratch*
  all Tier-1 clusters: scratch, projects

* You can also access staging storage through this endpoint (Leuven Tier-2 only)



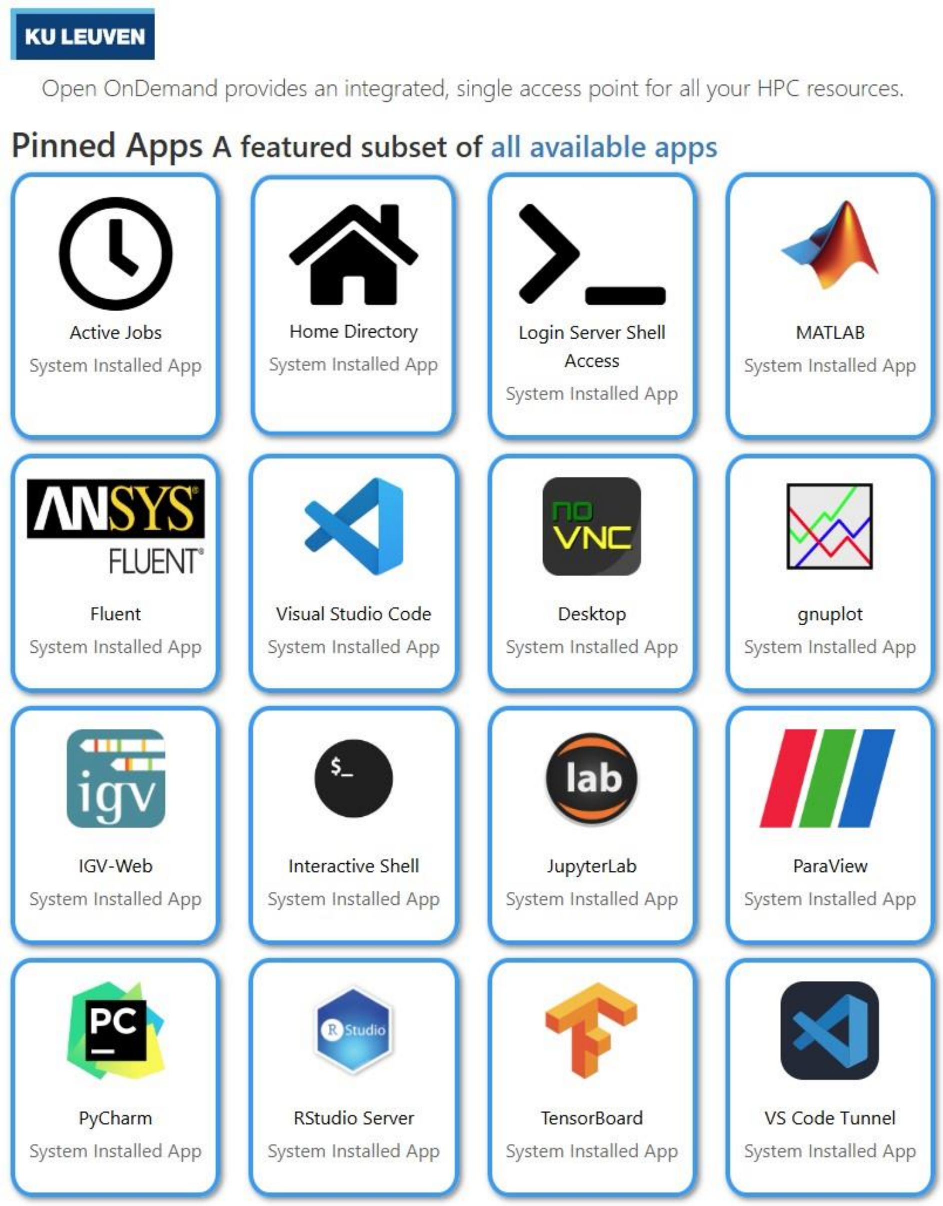

globus
globus.org
Share data
Transfer data
transfer
Secure endpoint
2 Globus moves the data
Secure endpoint
A
B
1 You submit a transfer request
3 Globus notifies you when the transfer is complete
2 Globus tracks shared files – no intermediate cloud storage!
Data source
1 User selects what to share and sets access permissions
3 User B logs in to Globus to access shared data

Open OnDemand

Open OnDemand

- Access clusters via your web browser
- https://ondemand.hpc.kuleuven.be
- Login via your institute's IDP
- File browser + Data Transfer (Globus)
- Develop, compile and test your code
- Interactive apps + integrated shell
- noVNC Desktop for graphical apps
- VSCode and PyCharm editors
- Fluent, Jupyter Lab, Rstudio, MATLAB, ParaView, Tensorboard, Gnuplot

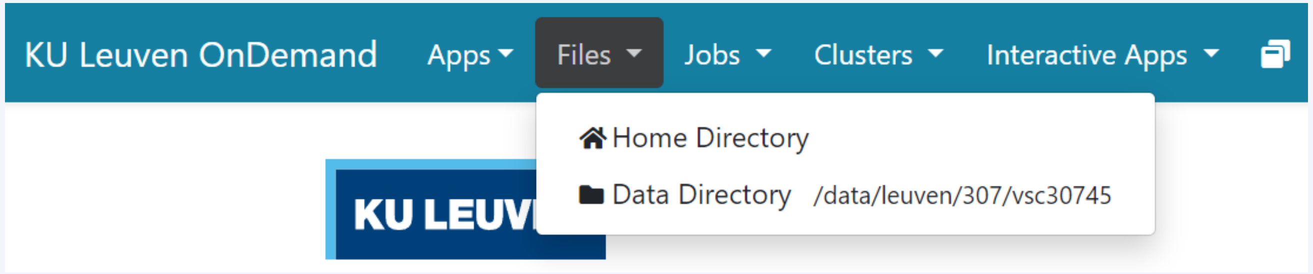
Please close your interactive sessions when you no longer actively use them!



Open OnDemand

Access your VSC storage

- Use the “Files” tab to access your `VSC_HOME` and `VSC_DATA`
- `VSC_SCRATCH` is not available from this menu

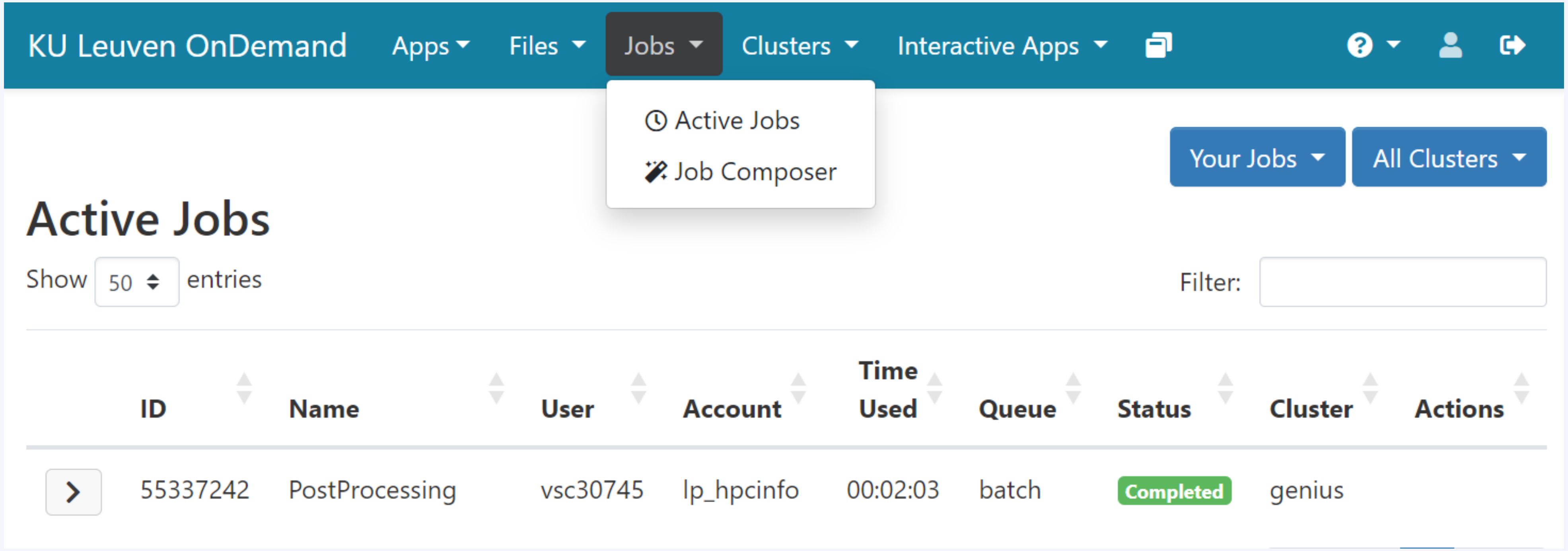


File/Folder management

- OK for transferring small files/folders
- Open any (sub)folder in terminal (will open a new tab)
- Deleted files/folders can be retrieved from (hourly, daily, weekly) snapshots

Open OnDemand

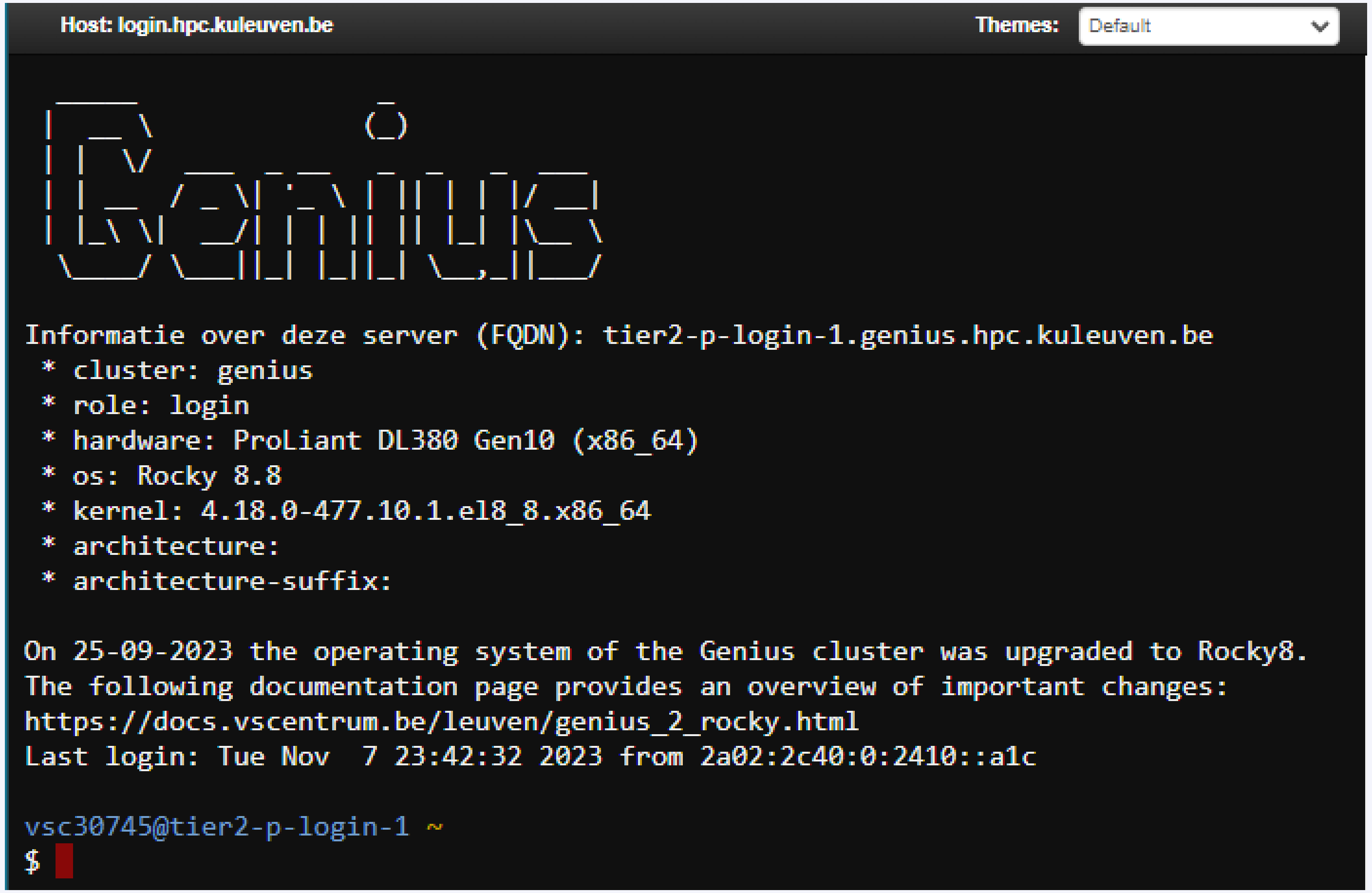
Monitor your active jobs



Open OnDemand: Login Server Shell Access

Login shell

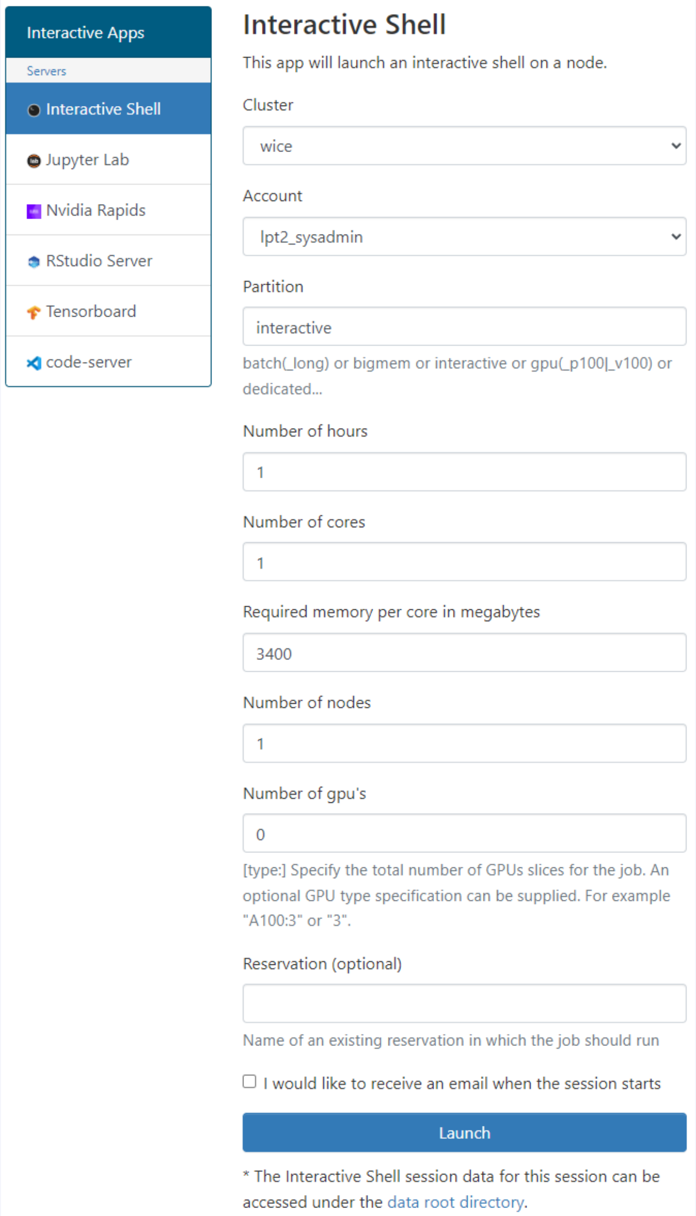
- Starts a shell in a new browser tab
- You land on a Genius login node (from which you can also access wICE)
- You land on your `VSC_HOME`. `cd` to data/scratch/staging immediately
- Do NOT compute on the login nodes
- Try:
  `$ sam-balance`
  `$ myquota`



Open OnDemand: Interactive Shell

Interactive Shell

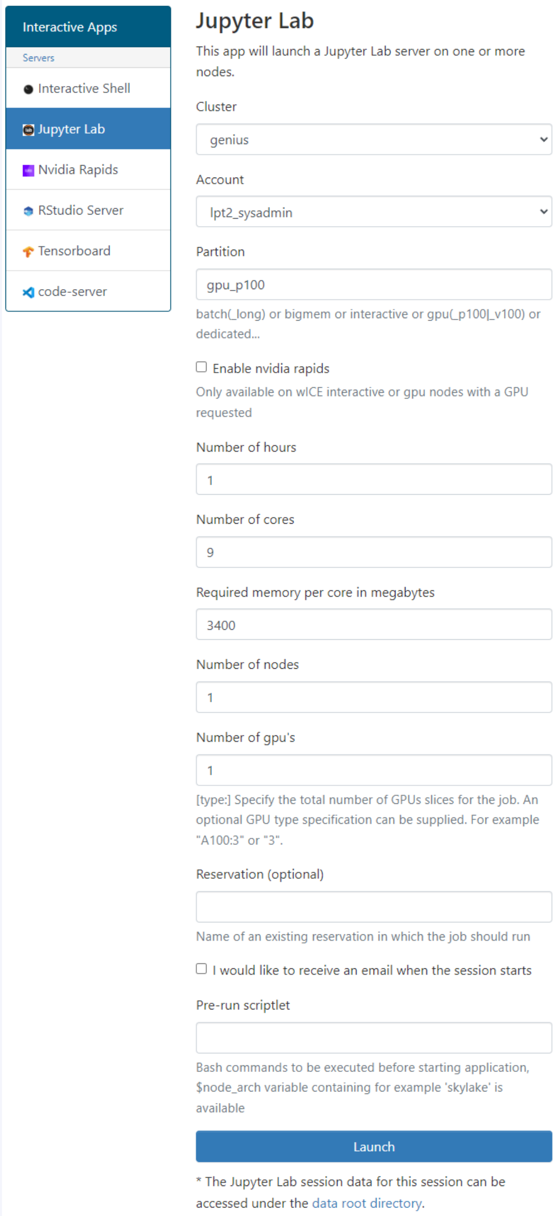
- Start a shell as a job (on a compute node)
- Recommended to:
  Install your own software (in `VSC_DATA`)
  Test/run your program interactively
  Debug a program
- Choose the relevant cluster, partition and resources (cores, memory, GPU)



Open OnDemand: JupyterLab

Jupyter Lab

- Create a notebook as a job (on a compute node)
- Suitable for:
  - pre/post-processing your data
  - prototyping
  - adding figures and text for pedagogical purposes
- Choose the appropriate cluster, partition and resources (cores, memory, GPU)
- User your own Python/R kernels (next slide)



Open OnDemand: Custom R/Python kernels

Custom kernels

- The default kernels may not offer the R/Python/… interpreter version or packages you want
- You can create a miniconda env and add it to your OnDemand kernels

Step 1: Miniconda

Start an Interactive Shell

Go to your data folder

```
$ cd ${VSC_DATA}
```

Download Miniconda

```
$ wget https://repo.continuum.io/miniconda/Miniconda3-latest-Linux-x86_64.sh
```

Install Miniconda in your VSC_DATA

```
$ bash Miniconda3-latest-Linux-x86_64.sh -b -p $VSC_DATA/miniconda3
```

Permanently add the path to Miniconda to your `~/.bashrc`

```
$ echo 'export PATH="${VSC_DATA}/miniconda3/bin:${PATH}" ' >> ~/.bashrc
```

Open OnDemand: Custom R/Python Kernels

Step 2: Environment setup

Create a new environment `<env_name>` with all packages you need, e.g.

```
$ conda create --name=<env_name> numpy scipy …
```

Enable your new environment

```
$ source activate <env_name>
```

Add the ipykernel package

```
$ conda install -c conda-forge ipykernel
```

Step 3: Register your kernel

```
$ python -m ipykernel install --user --env PYTHONPATH "" \
         --name <env_name> --display-name <env_name>
```

Step 4: Launching JupyterLab

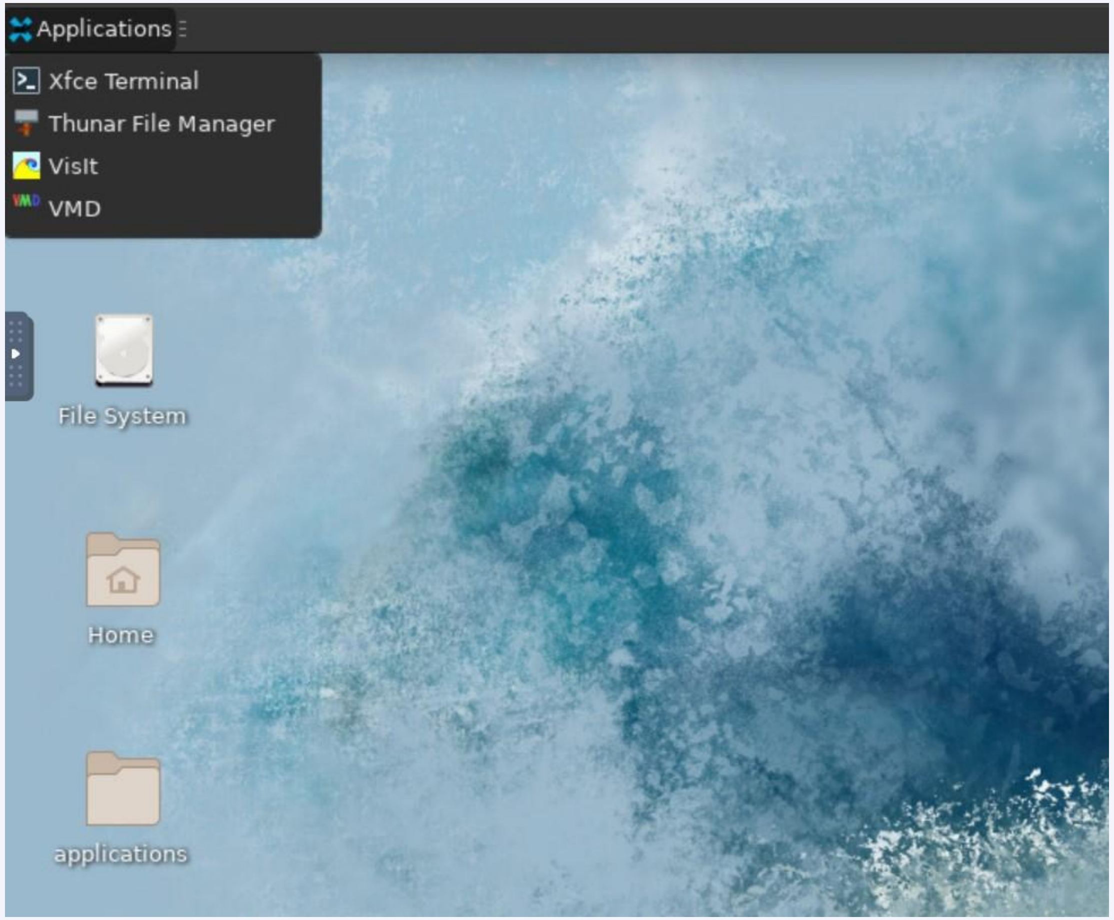
Now start a new Jupyter Lab session

Your new kernel should be available there

Open OnDemand: noVNC Desktop

Desktop

- Ideal for apps with GUI
- Load (graphical) modules in Terminal, e.g. MATLAB, Fluent, ncview, VisIt, …
- Minimalistic Terminal (different from login node)
- Two visualization apps (VisIt and VMD)



Open OnDemand: Other apps

code-server
System Installed App

- Start Visual Studio Code server as a job
- Develop, deploy and debug your workflow directly on HPC
- Use of the interactive partitions is free of charge and is a good choice
- Supports many languages + Github integration

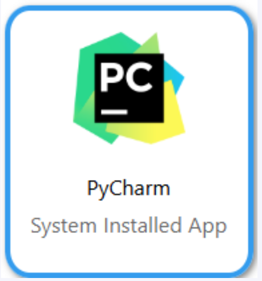
RStudio Server
System Installed App

- Start RStudio IDE as a job
- Install your own packages in VSC_DATA

Tensorboard
System Installed App

- Track machine learning metrics and visualize data during the workflow
- You need to provide a log folder

Open OnDemand: Other apps



Start PyCharm IDE as a job

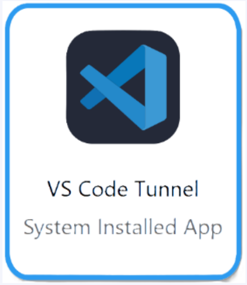
Develop, deploy and debug your Python workflow directly on HPC

Use Python environment based on Conda or virtual environments

Use Fluent for CFD simulations (single-node)

You need to be in a license group before being able to see the app logo

Contact your supervisor to help you joining an existing license group

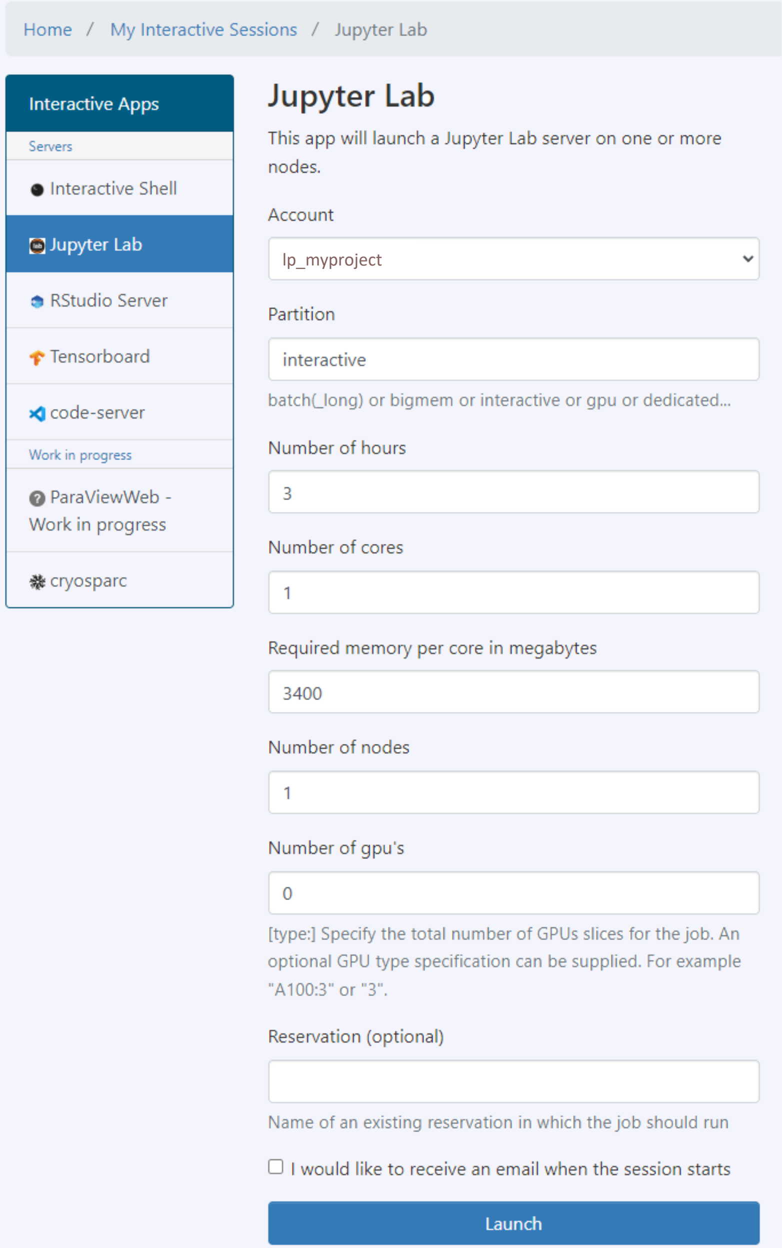


Tunnel to the cluster from your local VSCode, so that you code locally, but run remotely! You can enable extensions such as Copilot and more.

Open OnDemand: Resources

E.g. to start a Jupyter notebook

- Pick a valid Slurm credit account
- Default partition for most apps: *interactive* (interactive: max 8 cores, 1 GPU instance, 16 hr)
- Default resources: 1 core, 1 hour, 3400 MB RAM, no GPU



Software

Software: Centrally installed modules

- OS: wICE is running on Rocky Linux 9.6 and Genius (GPU nodes) migrate to that, soon.
- Toolchains:
  - *intel* (icc, icpc, ifort; Intel MPI; Intel MKL)
  - *foss* (gcc, g++, gfortran; OpenMPI; ScaLAPACK, OpenBLAS, FFTW)
  - Toolchain year on Genius from 2018a onwards; on wICE from 2021a onwards
- Note: not all toolchains are compatible with each other (conflicting dependencies)
  For example, **never mix …-*foss*-… and …-*intel*-…** modules

Command	Remark
`module av`	List all available modules
`module av Python`	List all Python-related modules
`module spider Python`	Get more info
`module load Python/3.12.3-GCCcore-13.3.0`	Load a specific module
`module list`	List all loaded modules and their dependencies
`module unload Python/3.12.3-GCCcore-13.3.0`	Unload a module (but dependencies still stay)
`module purge`	Remove all modules from your work session

Software: Your specific needs

- If you need a software package that is not yet available as a module, you can ask us to provide it
  - For Python and R packages, however, we prefer that you install it yourself
  - Read more about Python Package Management
  - Read more about R Package Management

- If you want to compile software yourself:
  - Best install it in your VSC_DATA
  - Use *intel* or *foss* toolchains
  - Compile your code on a compute node (with interactive job)
  - If you run into problems, don't hesitate to contact us

Starting to compute

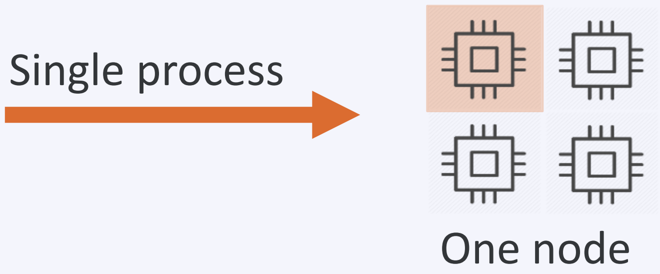
Resource glossary

- **Cluster:** which machine to use? Genius or wICE?
- **Nodes**: how many compute servers to allocate?
- **Cores**: how many cores per node to allocate?
- **Memory**: how much CPU memory to allocate?
- **Partition**: batch, gpu, bigmem, …?
- **Walltime**: how long to allocate the resources?
- **Storage**: how much storage (data, scratch, etc) will the job require?
- **Credits**: how many compute credits will the job consume?

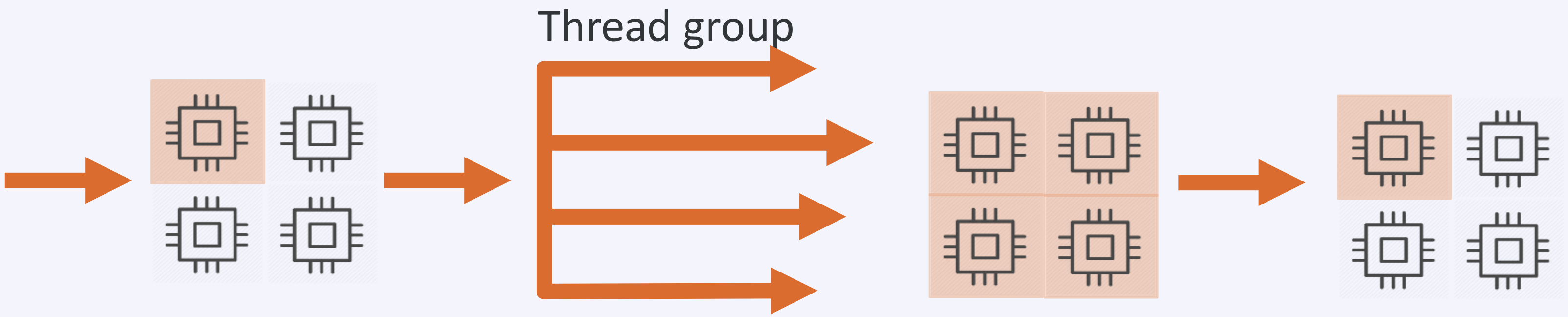
Default resources

- **Cluster:** the cluster associated with the node you’re on (Genius or wICE)
- **Nodes:** 1
- **Cores:** 1
- **Memory:** depends on which node(s) you use
- **Partition:** batch
- **Walltime:** typically 1 hour
- **Storage:** no default
- **Credits:** no default

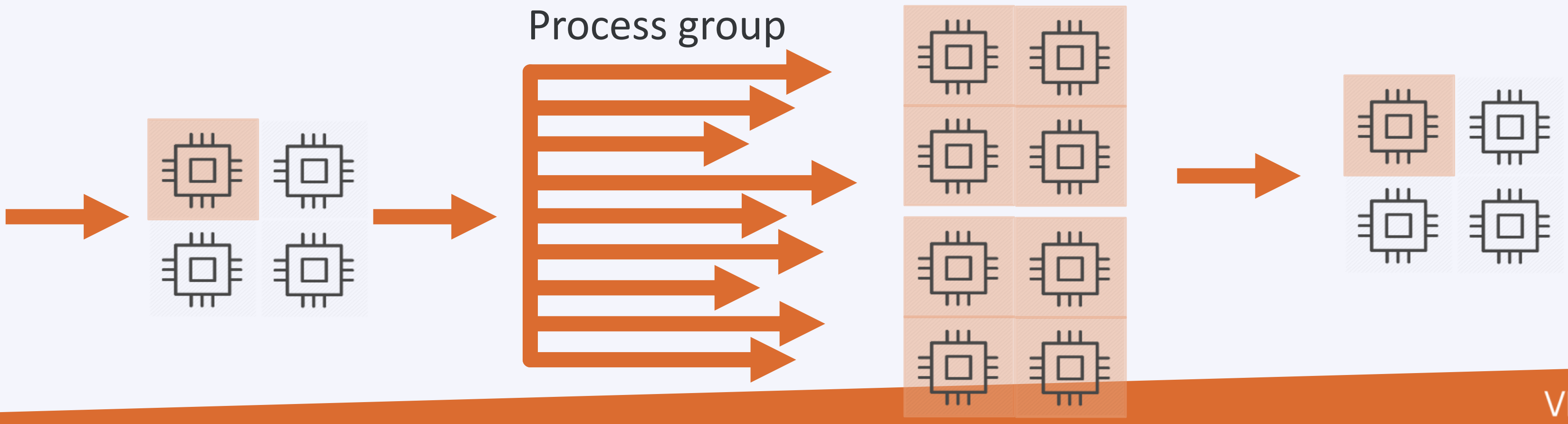
Serial application
1 process on 1 core



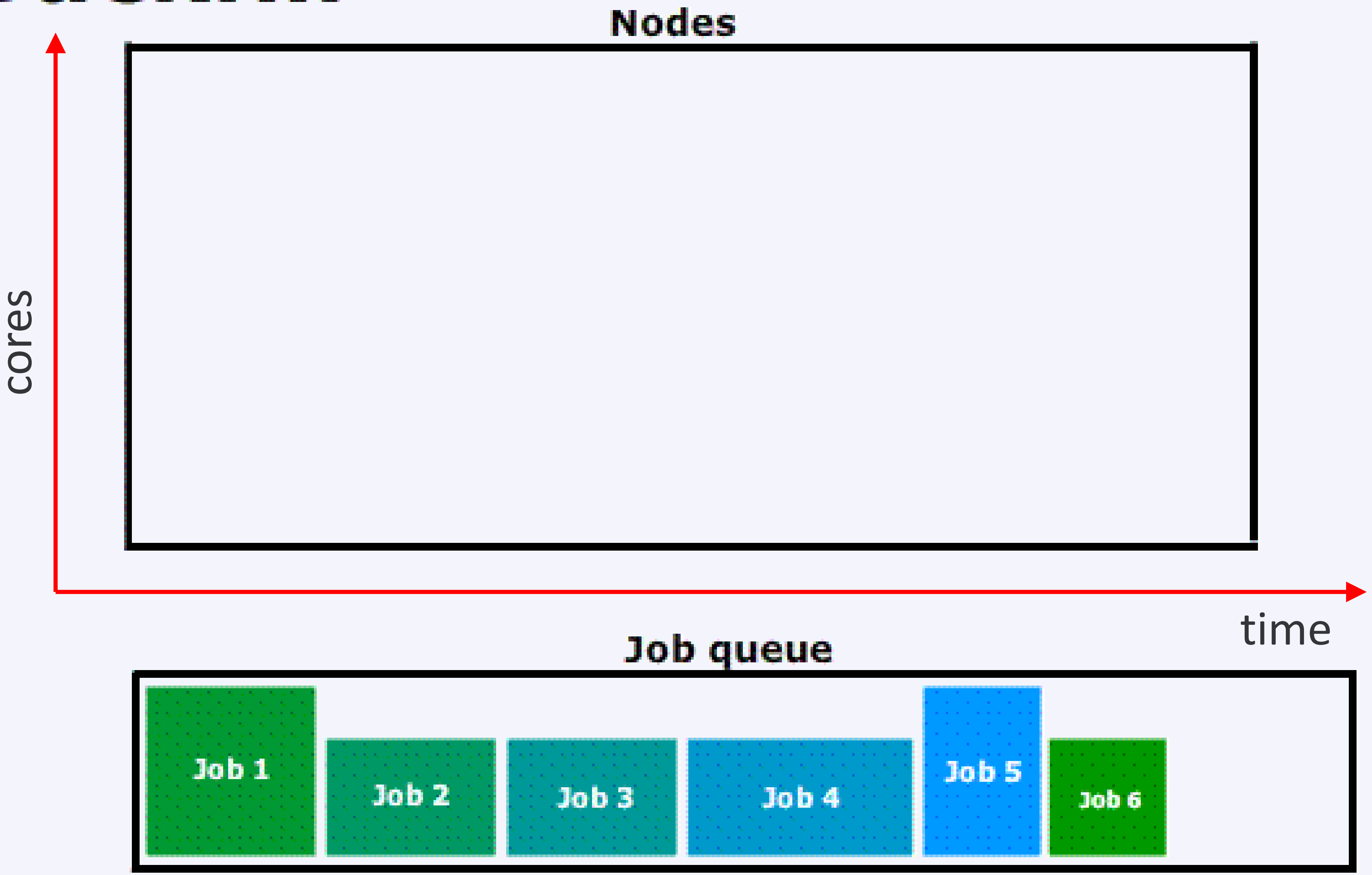
Multi-threaded application
N threads on N cores on **1 node**



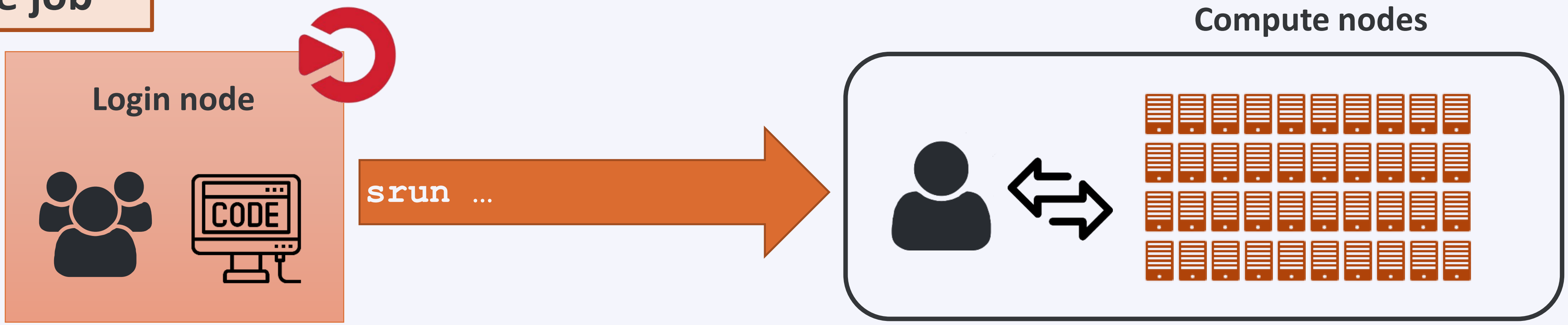
Distributed application
Many processes on multiple cores, possibly spread over multiple nodes



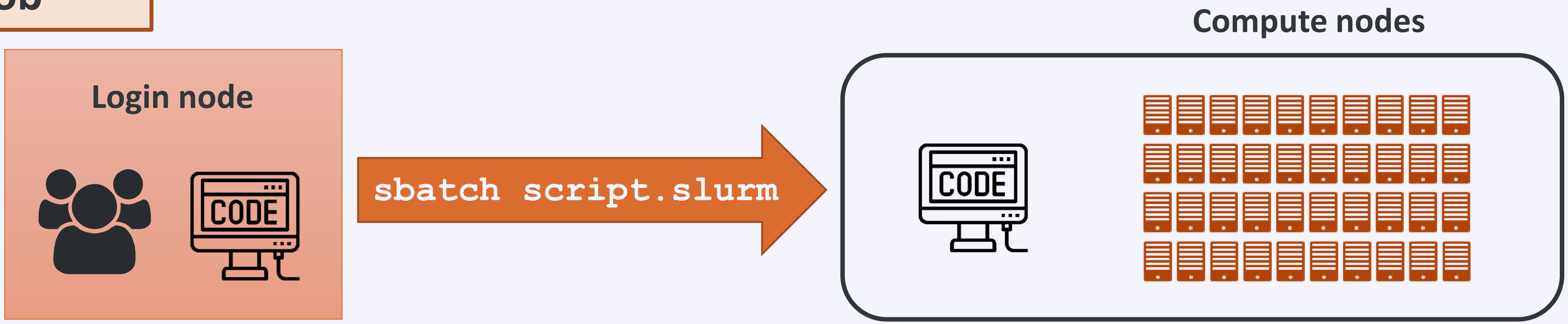
Backfill



Interactive job



Batch job



Example interactive job on **wICE**

Request an interactive job

Credit account for compute credits

For special hardware or queue

Target cluster

```
$ srun --account=<project name> --partition=interactive --clusters=wice  \
    --nodes=1 --ntasks=4 --time=01:00:00 --pty bash -l
```

Number of nodes

Number of processes or cores

Max. runtime

Interpret with Bash

Remark

- Specifying <project name> for credits is mandatory, *e.g.* **`-A lp_hpcinfo`**
- Implicit defaults for an interactive job on wICE are
  `-n 1 -t 01:00:00 -p batch --mem-per-cpu=3400M`

Slurm command-line tools

Command	Purpose
`$ sbatch ...`	Submit a batch job
`$ srun ...`	Submit an interactive job
`$ scancel --cluster=wice <JobID>`	Cancel a specific pending or running job
`$ scontrol show job --clusters=wice <JobID>` `$ slurm_jobinfo <JobID>`	Detailed job info (very useful to diagnose issues)
`$ squeue --clusters=wice --me --long`	Status of all recent jobs
`$ squeue --clusters=wice --me --start`	Give a rough estimate of when your jobs will start
`$ sinfo --clusters=wice`	Info about the state of available partitions and nodes
`$ sacct --clusters=wice --batch --job <JobID>`	Show the jobscript used for a given batch job
`$ slurmtop`	Overview of the cluster
`$ sam-balance`	Overview of all your available credit projects
`$ sam-list-allocations`	Detailed overview of your credit allocation history

Slurm job submit options

Slurm option	Remarks

`--nodes=X --ntasks-per-node=Y`
`--ntasks=X*Y`
`--ntasks=X --cpus-per-task=Y`
	or
	or
	(*e.g.* Hybrid MPI + OpenMP)

`--partition=<partition_name>`	Default: "batch" partition
`--mem-per-cpu=<size><M>`	Min memory per core, e.g. 5000M (Genius)
`--time=<dd-hh:mm:ss>`	e.g. 1-12:30:00
`--job-name=<job_name>`	
`--output=<file_template>`	STDOUT; defaullt="slurm-%j.out"
`--error=<file_template>`	Default: redirect to STDOUT
`--mail-type=FAIL,BEGIN,END`	
`--mail-user=<email-address>`	
`--export=<ALL,key=value>`	Additional variables to pass to the job
`--account=<account_name>`	Credit account (mandatory)

Slurm job submit options

Slurm option	Remarks
--account=<account_name>	Mandatory (credits)
--partition=<partition_name>	Default: "batch" partition
--time=<dd-hh:mm:ss>	e.g. 1-12:30:00
--nodes=X --ntasks-per-node=Y --ntasks=X*Y --ntasks=X --cpus-per-task=Y	or or (*e.g.* Hybrid MPI + OpenMP)
--mem-per-cpu=<size><M>	Min memory per core, e.g. 5000M (Genius)
--job-name=<job_name>	
--output=<file_template>	STDOUT+STDERR; defaullt="slurm-%j.out"
--mail-type=FAIL,BEGIN,END	
--mail-user=<email-address>	
--export=<ALL,key=value>	Additional variables to pass to the job

Long and short options

Some (not all) of the `sbatch`, `srun` and `salloc` command line options also have a short version

Short version	Long version	Meaning
`-A`	`--account`	Slurm account name
`-a`	`--array`	Job array range
`-M`	`--clusters`	Cluster name
`-c`	`--cpus-per-task`	Number of cores per task (default: 1)
`-d`	`--dependency`	Job dependency (after, afterok, afterany, …)
`-i`	`--input`	STDIN filename
`-e`	`--error`	STDERR filename
`-o`	`--output`	STDOUT filename
`-t`	`--time`	Maximum walltime
`-N`	`--nodes`	Minimum num nodes
`-n`	`--ntasks`	Maximum num tasks
`-p`	`--partition`	Partition name

Interactive jobs

- Interactive job with default options (1 core for 1 hour on the *batch* partition)

```
$ srun --clusters=wice --account=lp_hpcinfo --pty bash -l
```

- Interactive job with X-forwarding

```
$ srun --clusters=wice --account=lp_hpcinfo --x11 --pty bash -l
```

- Interactive job with multiple cores, on the *interactive* partition

```
$ srun --clusters=wice --account=lp_hpcinfo --nodes=1 --ntasks=4 \
       --partition=interactive --pty bash -l
```

- Interactive job with multiple cores and a GPU device

```
$ srun --clusters=wice --account=lp_hpcinfo --nodes=1 --ntasks=18 \
       --partition=gpu --gpus-per-node=1 --pty bash -l
```

- To join one of your running jobs (connecting to the first compute node in the job’s allocation):

```
$ srun -M wice --jobid=<JobID> --overlap -pty bash -l
```

Example Slurm job script

jobscript.slurm

```
#!/bin/bash -l
#SBATCH --clusters=wice
#SBATCH --account=lp_hpcinfo
#SBATCH --nodes=1
#SBATCH --ntasks=4
#SBATCH --mem-per-cpu=3400M
#SBATCH --job-name=hpc_workflow
#SBATCH --mail-type=BEGIN,END,FAIL
#SBATCH --mail-user=my.name@kuleuven.be

module load intel/2024a
module load Python/3.12.3-GCCcore-13.3.0
which python
cd $VSC_SCRATCH/projects/simulations
cp -r $VSC_DATA/input_data .

python modelling.py

cp -r output_data $VSC_DATA
rm -rf ./input_data ./output_data
```

Shebang

Resource list

Loading modules

Move data

Execute commands

Move data

Batch job workflow

- Submit the job to the scheduler
- Receive a unique JobID
- Error and output files

Job script

```
#!/bin/bash -l
…
```

Submit command

```
$ sbatch simulation.slurm
```

JobID

```
Submitted batch job 60042478 on cluster wice
```

stderr, stdout

```
$ ls *.out
slurm-60042478.out
```

Output files

- STDERR and STDOUT are by default redirected to a single file:

  `slurm-<JobID>.out`

- Contains job info, all errors and warnings, and printouts
- Worth going through (especially if it's a new kind of job)
- Address all warnings and errors (if you can)
- Typical error examples on the right

STDOUT + STDERR

```
$ ls slurm-*.out
slurm-60042478.out
```

Out of memory

```
slurmstepd: error: Detected 1 oom-kill event(s).
Some of your processes may have been killed by the
cgroup out-of-memory handler.
```

Short walltime

```
slurmstepd: error: *** JOB 60042478 ON s28c11n2
CANCELLED AT 2023-02-08T10:03:43 DUE TO TIME LIMIT
***
```

Low disk space

```
IOError: [Errno 122] Disk quota exceeded
```

Output files

- Standard output and error channels can be redirected to other files:
  `#SBATCH --output …`
  `#SBATCH --error …`

stdout

```
$ ls slurm-*.out
slurm-60041238.out
```

Output file

```
SLURM_JOB_ID: 60033947
SLURM_JOB_USER: vscXXXXX
SLURM_JOB_ACCOUNT: lp_wice_pilot
SLURM_JOB_NAME: testjob
SLURM_CLUSTER_NAME: wice
SLURM_JOB_PARTITION: batch
SLURM_NNODES: 1
SLURM_NODELIST: m28c30n4
SLURM_JOB_CPUS_PER_NODE: 72
Date: Tue Jan 10 17:02:04 CET 2023
Walltime: 00-01:00:00
==================================
/apps/leuven/rocky8/icelake/2021a/
softwar
e/intel-compilers/2021.2.0/compile
r/2021.2.0/linux/bin/intel64/icc
cp: cannot stat '/apps/leuven/trai
ning/test': No such file ordirecto
ry
Hello World
```

Resource summary

stdout

Job monitoring

Example

```
$ scontrol show job --clusters=wice 60049330
   JobId=60049330 JobName=f70.slurm
   UserId=vsc3…(253…) GroupId=vsc3…(253…)
   Account=lp_my_project QOS=lp_my_project
   JobState=PENDING Reason=QOSGrpBillingMinutes
   SubmitTime=2023-02-16T00:24:58 EligibleTime=2023-02-16T00:24:58
   Partition=batch NodeList= NumNodes=4-4 NumCPUs=288 NumTasks=288 CPUs/Task=1
   TRES=cpu=288,mem=979200M,node=4,billing=733
   MinCPUsNode=72 MinMemoryCPU=3400M
   WorkDir=/vsc-hard-mounts/leuven-data/3../vsc3…/Odonate_SOM
   StdErr=/vsc-hard-mounts/leuven-data/3../vsc3…/Odonate_SOM/slurm-60049330.out
   StdIn=/dev/null
   StdOut=/vsc-hard-mounts/leuven-data/3../vsc3…/Odonate_SOM/slurm-60049330.out
```

Diagnosis

Why does my job remain in a pending state?
Check out the “Reason” on Slurm docs: https://slurm.schedmd.com/resource_limits.html

Other wICE partitions

GPU

```
#SBATCH --partition=gpu
#SBATCH --nodes=1
#SBATCH --ntasks=18
#SBATCH --gpus-per-node=1
```

Big memory

```
#SBATCH --partition=bigmem
#SBATCH --nodes=1
#SBATCH --tasks-per-node=72
#SBATCH --mem-per-cpu=28000M
```

Interactive partitions

- Accessible via command line and Open OnDemand
- To quickly compile, test, debug your (parallel) application
- To pre- or postprocess your data (incl. visualizations)
- Short queue time
- 12 nodes on Genius: 36 cores/node, 192 GB mem
- 4 nodes on wICE: 64 cores/node, 512 GB mem, 1 physical GPU (split in 7 instances)
- Max resource per user: 8 cores, 1 GPU instance, 16 hour walltime
- Free of charge

Interactive job

```
$ srun --account=lp_myproject --clusters=wice \
       --partition=interactive --nodes=1 --time=16:00:00 \
       --gpus-per-node=1  --pty bash -l
```

Debugging partitions

- Accessible via command line and Open OnDemand
- Quickly test if your (parallel) application works
- Short queue time
- Only one job at a time, max walltime: 1 hr

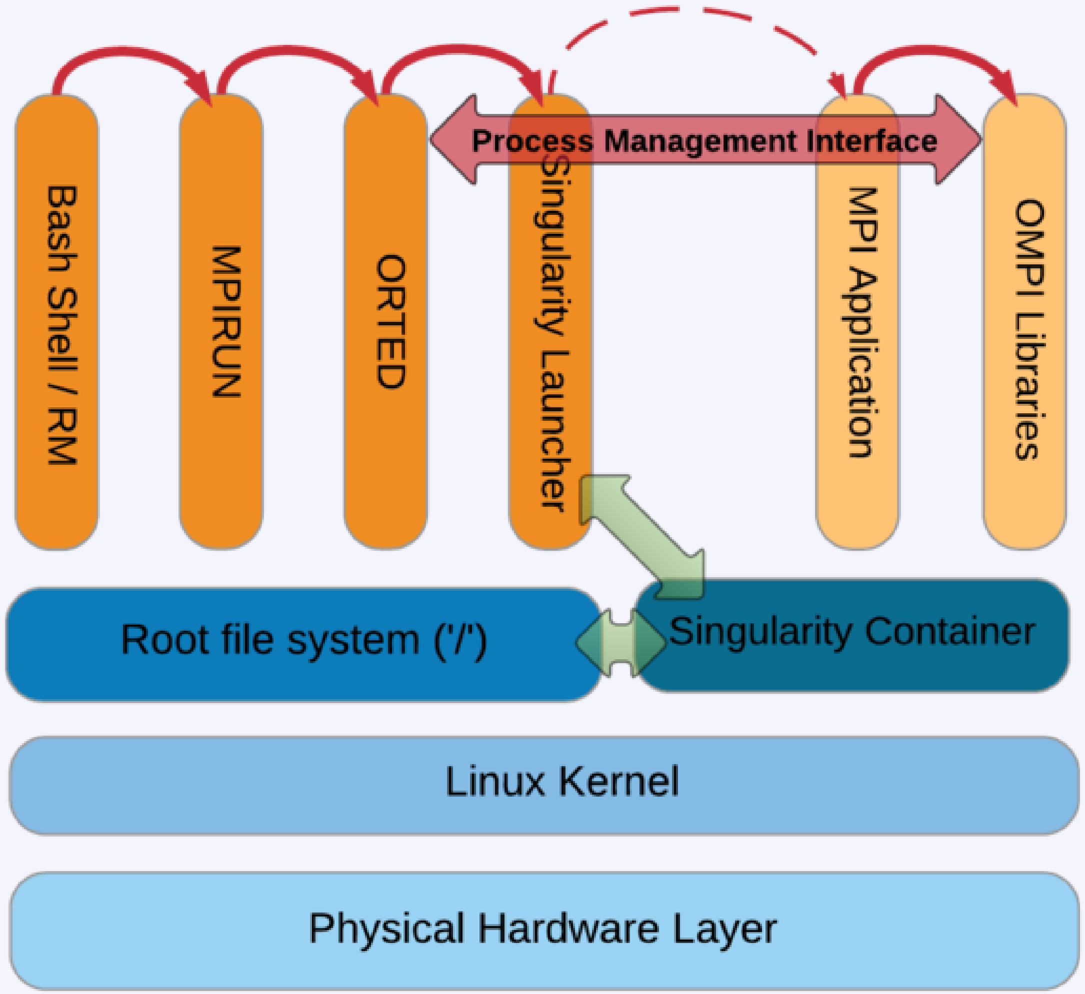
Cluster	Partition	Resources
Genius	batch_debug	2 Cascadelake nodes
	gpu_p100_debug	1 Skylake node with 4 P100 GPUs
wICE	interactive	4 Icelake nodes, each with 7 MIG instances of 1 A100 GPU
	gpu_a100_debug	1 Icelake node with 1 A100 GPU

Containers: Apptainer (Singularity)

- What?
  - Self-contained OS image
    in which you can install software and include data
- Why?
  - Package managers like `apt` can be convenient
  - Portability: images can be used on various hardware (but performance may be sub-par)
- How?
  - Create the image
  - Run it on wICE or Genius
  - Parallelization should work
    (multi-node MPI runs may require some care)

script.slurm

```
#!/bin/bash -l
#SBATCH --time=30:00
#SBATCH --nodes=1
#SBATCH --ntasks=72
#SBATCH --clusters=wice
#SBATCH --account=lp_hpcinfo
apptainer run Project.sif ./model.exe
```



High-throughput computing with *atools*

- You can use `atools` to handle high-throughput computing workloads
- `module load atools/<version>`
- VSC training: Workflows for Data Science
- Training material: https://github.com/gjbex/Workflows-for-HPC
- Recording: https://www.youtube.com/watch?v=GlwaRyi1LVA
- User documentation: https://atools.readthedocs.io/en/latest/

Demo: try things yourself

Demo: try things yourself

- Request to become a member of the *lp_hpcinfo* group via account.vscentrum.be
- Transfer a file with FileZilla
- Check your disk quota
- Check your available credits
- Check/load/list/unload/purge module

Demo: try things yourself

- Copy `/apps/leuven/training/HPC_intro/` to your `$VSC_DATA`
- Submit `cpujob.slurm` to the cluster
- List all your jobs with `squeue -M wice`
- Check the information about the job with `slurm_jobinfo -M wice <job_ID>`
- Modify the `mpi.slurm` script to request 1 node, 72 cores for 30 minutes, with job start/end e-mail notifications
- Check the status of all the jobs

Demo: job monitoring

- Submit an interactive job
  - Run your program on a compute node
  - Open a new terminal and `ssh` to the allocated compute node
  - Check the resources usage with `htop`
- Submit a GPU batch job
  - While the job is running, check its details with `slurm_jobinfo`
  - and check resource on the compute node using `ssh`, `htop` and `nvidia-smi`

VLAAMS SUPERCOMPUTER CENTRUM

Useful links

Helpdesk (email): hpcinfo@kuleuven.be

Helpdesk (online form): https://admin.kuleuven.be/icts/HPCinfo_form/HPC-info-formulier

VSC website: https://www.vscentrum.be

VSC documentation: https://docs.vscentrum.be

VSC agenda, training sessions, events (User Day): https://www.vscentrum.be/vsctraining

System status page: https://status.vscentrum.be

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